



Creditor rights, information sharing, and bank risk taking[☆]

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ABSTRACT

Looking at a sample of nearly 2,400 banks in 69 countries, we find that stronger creditor rights tend to promote greater bank risk taking. Consistent with this finding, we also show that stronger creditor rights increase the likelihood of financial crisis. On the plus side, we find that stronger creditor rights are associated with higher growth. In contrast, we find that the benefits of information sharing among creditors appear to be universally positive. Greater information sharing leads to higher bank profitability, lower bank risk, a reduced likelihood of financial crisis, and higher economic growth.

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1. Introduction

A growing literature dating back to La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998, 1999) demonstrates the important connections between legal systems, investor protections, and the development of capital markets. Research in this area generally finds that stronger investor protections help promote capital market development, which in turn is often correlated with economic development. These important insights have spurred further exploration into the various channels in

which the legal system influences the formation of capital and economic growth.

In this spirit, a number of more recent papers have specifically examined both the effects of stronger creditor rights in bankruptcy, and the role of information sharing among creditors. In a study of 129 countries over a 25-year time period, Djankov, McLiesh, and Shleifer (DMS, 2007) find that the ratio of private credit to gross domestic product is positively related to stronger creditor rights, stronger legal protections, and information sharing among creditors. John, Litov, and Yeung (2008) find that stronger corporate governance is correlated with greater corporate risk taking. At the same time, Acharya, Amihud, and Litov (2009) find that stronger creditor rights lead to reduced corporate risk taking in the form of diversifying acquisitions. Claessens and Klapper (2005) find that the various components of the popular creditor rights indexes have a differential effect on the likelihood of bankruptcy, while Brockman and Unlu (2009) find that companies are

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less likely to pay dividends in countries with weaker creditor rights. Another strand of the literature has examined the role of credit information sharing in enhancing credit availability (Pagano and Jappelli, 1993; Padilla and Pagano, 1997; Djankov, McLiesh, and Shleifer, 2007; Brown, Jappelli, and Pagano, 2009), in lowering the cost of credit (Brown, Jappelli, and Pagano, 2009), in reducing default rates (Jappelli and Pagano, 2002), and in corruption in lending (Barth, Lin, Lin, and Song, 2009).

Noticeably absent in this recent literature is an examination of the links between creditor rights, information sharing, and bank risk taking. Perhaps the most closely related paper is the recent work by Cole and Turk-Ariss (2008) who find that banks have higher loan ratios in countries with English legal origin and weaker creditor rights. The limited research in this area is somewhat surprising given the importance of bank lending in promoting economic development (see, for example, Beck, Levine, and Loayza, 2000; Levine, 2005), and given the impact that creditor rights and information sharing are likely to have on both the relative importance of bank lending and the level of bank risk taking across different countries.¹ Arguably, the recent global financial crisis has further intensified interest in understanding the various factors that affect both bank risk taking and the likelihood of financial crisis.

This paper attempts to fill this gap by exploring in detail the interactions between the level of creditor rights, information sharing, and risk taking among banks in 69 countries. We further explore how creditor rights and information sharing affect both the likelihood of a financial crisis in the country's banking system and the level of industry growth. Most notably, we find that stronger creditor rights are correlated with higher bank risk taking. We also find that information sharing among creditors reduces bank risk taking, and also has an interactive effect which mitigates the effect that creditor rights have on bank risk taking. These conclusions are extremely robust with respect to alternative measures of risk taking, different control variables, and different estimation techniques. Moreover, we find that stronger creditor rights increase the likelihood of financial crisis, whereas information sharing reduces the likelihood of crisis. Finally, we observe that stronger creditor rights and information sharing are both correlated with higher industry growth.

Our study controls for bank characteristics, country characteristics, a variety of measures of investor protection used in the law and finance literature, as well as the structure of national bank regulations. In this regard, our study complements a number of recent papers including Laeven and Levine (2009) who find that ownership structure and national bank regulations influence bank risk taking, Qian and Strahan (2007) who find that creditor protection influences the pricing and structure of bank loan contracts, and Kroszner, Laeven, and Klingebiel (2007) who find that banking crises have a

more pronounced effect on sectors that are heavily dependent on external finance.

While our study is not able to directly answer the larger questions regarding optimal risk taking, our findings provide interesting insights into the channels in which creditor rights and information sharing among creditors influence bank lending decisions, which in turn affect the level of growth and the likelihood of financial crisis. In particular, our results suggest that there is both a “bright” side and “dark” side to enhanced creditor protections. Simply put, stronger creditor rights appear to encourage banks to take on more risks which helps provide valuable capital to private firms which enhances overall economic growth. On the downside, the “dark” side of greater risk taking is that it significantly increases the likelihood of financial crisis. In this respect, our findings complement the recent work of Rancière, Tornell, and Westermann (2008) who look more broadly at the connection between systemic crises and growth. While their study does not consider the effects of creditor rights (or information sharing), they do demonstrate that countries that have experienced a financial crisis tend to have higher growth rates relative to countries with more stable financial conditions.

In contrast to these conflicting effects regarding creditor rights protections, information sharing among creditors appears to be universally beneficial. Specifically, information sharing reduces information asymmetries and enhances transparency—which reduces bank risk taking and the likelihood of crisis, while at the same time promoting economic growth.

This paper contributes to the literature in at least three important ways. First, we add to the law and finance literature by further demonstrating the significant and interactive effects of creditor rights and information sharing. Far from having a neutral effect, we show that these institutional features have a profound influence on the level of bank risk taking, the level of economic growth, and the likelihood of financial crisis. Second, our research contributes to the literature that explores the determinants of bank risk taking (e.g., Saunders, Strock, and Travlos, 1990; Houston and James, 1995; Laeven and Levine, 2009). Given the significance of bank lending (particularly in developing countries), it is important to understand how the legal, regulatory, and institutional environment influences banks' willingness to lend and take risk. Third, we add to the literature that focuses on the determinants of financial crisis (e.g., Demirgüç-Kunt and Detragiache, 2002; Beck, Demirgüç-Kunt, and Levine, 2006; Kroszner, Laeven, and Klingebiel, 2007). Here we find compelling evidence that stronger creditor rights increase the likelihood of financial crisis, whereas information sharing reduces the likelihood of financial crisis. While these issues are always important, there is obviously renewed interest in these questions given the magnitude of the recent global financial crisis.

The rest of the paper proceeds as follows. Section 2 reviews the various components of creditor rights and information sharing and discusses the various channels in which they are likely to affect bank risk taking. Section 3 describes the data and presents summary statistics.

¹ Several recent theoretical studies look at the effects of competition on bank risk taking, with mixed predictions. See, e.g., Allen and Gale (2000) and Boyd and De Nicolò (2005).

Section 4 reports the key results relating bank risk taking to measures of creditor rights and information sharing, while Section 5 reports the results that relate these measures to the likelihood of financial crisis and the level of industry growth. Section 6 concludes.

2. Creditor rights, information sharing, and incentives for bank risk taking

We hypothesize that the strength of creditor rights and the level of information sharing among creditors are likely to have important influences on the level of bank risk taking. These influences are also expected to affect the likelihood of financial crisis, and banks' willingness to provide credit, which will in turn affect the level of economic growth. In this section, we review the various components of creditor rights and information sharing and discuss their potential effects on bank risk-taking.

2.1. Measuring creditor rights

Following LLSV (1998), we use the creditor rights index to measure the powers of secured creditors in bankruptcy. This index consists of four components: (1) whether there are restrictions imposed, such as creditors' consent or minimum dividend, when a debtor files for reorganization (*Restrictions on reorganization*); (2) whether secured creditors are able to gain possession of assets after the petition for reorganization is approved, that is, whether there is no automatic stay or asset freeze imposed by the court on a creditor's ability to seize collateral (*No automatic stay*); (3) whether secured creditors are ranked first in the distribution of proceeds of liquidating a bankrupt firm as opposed to other creditors such as government or workers (*Secured creditor paid first*); and (4) whether the incumbent management does not stay in control of the firm during the reorganization, in other words, whether an administrator, not the management, is responsible for running the business during the reorganization (*No management stay*). A value of one is added to the index when a country's laws and regulations provide each of these powers to secured lenders. The aggregate creditor rights index (*Creditor rights*) therefore ranges from zero to four with a higher value indicating stronger creditor rights. Djankov, McLiesh, and Shleifer (2007) extend the creditor rights data set to include annual observations across 129 countries over the period from 1978 to 2003. In our analysis, we use both the aggregate index and the four components in the extended sample to explore the relationship between creditor rights and bank risk taking.

2.2. Information sharing among creditors

The level of information sharing among creditors is also likely to have an important influence on bank risk taking. Based on the data available from DMS (2007) and the World Bank "Doing Business" data set, we include various variables to measure information sharing among lenders. Following DMS (2007), the first variable (*Information sharing*) indicates whether an information

sharing agency (public registry or private bureau) exists. This variable equals one if an information sharing agency is operating in the country by the end of 2005, and zero otherwise. Both public registry and private bureau are databases maintained by a public authority or private commercial firm, which collect information on the creditworthiness of borrowers in the financial system and facilitate the exchange of credit information among banks and financial institutions (DMS, 2007). The depth of credit information, however, varies across countries and regions. Some agencies only collect limited information on outstanding loans of large borrowers, while other agencies distribute extensive information including late payments and defaults, demographic data, credit inquiries, ratings, sometimes even the payment of utility bills, and court records of the company and its owners (Miller, 2003; DMS, 2007). We therefore use the second variable (*Depth of credit information*) to capture the difference in information contents across countries. The data are from the World Bank "Doing Business" data set. Specifically, the depth of credit information index measures rules affecting the scope, accessibility, and quality of credit information available through either public or private credit registries. The six characteristics measured by the index include: (1) both positive credit information (for example, loan amounts and pattern of on-time repayments) and negative information (for example, late payments, number and amount of defaults, and bankruptcies) are distributed; (2) data on both firms and individual borrowers are distributed; (3) data from retailers, trade creditors, or utilities, as well as from financial institutions, are distributed; (4) more than two years of historical data are distributed; (5) data are collected on all loans of value above 1% of income per capita; and (6) laws are provided for borrowers' rights to inspect their own data. A value of one is added to the index when a country's information agencies have each of these characteristics. The index ranges from zero to six, with higher values indicating the availability of more credit information, from either a public registry or a private bureau, to facilitate lending decisions.

2.3. Creditor rights, information sharing, and bank risk taking

Creditors and stockholders often have conflicting incentives. Perhaps most notably, stockholders have greater incentives for risk taking, but rational creditors will consider these incentives when deciding the conditions under which to grant credit. Consequently, it is often incumbent upon borrowers to find ways to credibly commit that they will not engage in excessive risk taking. To the extent that the strength of creditor rights and the degree of information sharing affect the contracting environment, they are also likely to have important influences on creditors' ex ante incentives as well as the recovery rates in the event of a default.

The levels of creditor rights and information sharing are also likely to influence the overall level of credit. DMS (2007) hypothesize that there are two channels in which

stronger creditor rights and information sharing promote private lending. One possibility is that stronger creditor rights grant more power to creditors in bankruptcy, which makes them more willing to grant credit. The second possibility is that information sharing among creditors helps reduce costly information asymmetries, which enable lenders to provide more credit.²

For a given set of borrowers, we might expect that, all else equal, stronger creditor rights would translate into lower bank risk taking. Specifically, in an environment with stronger creditor rights, lenders are more likely to grab collateral, force repayment, or even gain control of the debtor that is in financial distress. It follows that stronger creditor rights would lead to higher recovery rates in the event of default, which reduces the creditor's risk. Moreover, stronger creditor rights, such as restrictions on reorganization and forced change of management during the reorganization process, clearly have negative consequences for the firm's management if the firm enters financial distress.³ These facts suggest that borrowers are less willing to take risks when creditors are better protected. Supporting this hypothesis, [Acharya, Amihud, and Litov \(2009\)](#) find international evidence that stronger creditor rights tend to reduce corporate risk taking.

However, stronger creditor rights may also induce banks to provide loans to a wider (potentially riskier) set of borrowers. Such a pattern is consistent with [DMS \(2007\)](#) finding that stronger creditor rights are correlated with increased lending. In particular, with greater protection in the event of a default, creditors may be more willing to lend to riskier borrowers with poorer credit ratings. This effect will increase the average expected default rate in the bank's portfolio. If the higher expected recovery rates in default fail to offset the higher expected default rates, stronger creditor rights would be associated with increased bank risk.

These two channels suggest that stronger creditor rights influence the relative supply and demand for credit, which in turn may affect the level of interest rates, the terms of contracting, and the competitive position of banks relative to other suppliers of credit—each of which may indirectly have further implications for the level of risk taking. More specifically, stronger creditor rights are likely to increase expected recovery rates, which may make banks more willing to provide credit to a wider set of borrowers, thereby increasing the supply of credit. At the same time, as mentioned above, stronger creditor rights may increase the firm and management's costs associated with bankruptcy, which induces some borrowers to seek other sources of funding and leads to a

decrease in the demand for credit. Consistent with this argument, [Acharya, Amihud, and Litov \(2009\)](#) find that companies have lower financial leverage in countries with stronger creditor rights.⁴ They also find evidence that strengthening a country's creditor rights lowers firms' financial leverage. Moreover, in addition to reducing borrowers' demand for debt, there may also be “substitution effects” influencing the borrower's preferred mix of bank credit, public debt, and trade credit. With better protection, bondholders are less worried about the problems of firm managers' opportunistic behavior and the controlling shareholders' expropriation ([Jensen and Meckling, 1976](#); [Klock, Mansi, and Maxwell, 2005](#)). They are thus more willing to buy public bonds issued by corporations. Indeed, [Boubakri and Ghouma \(2008\)](#) find international evidence that greater creditor rights protection reduces bond costs and increases corporate bond ratings. The resulting relative attractiveness of public debt may reduce firms' demand for bank loans, which would carry more restrictive conditions (more covenants, e.g.).⁵ At the same time, stronger creditor rights may induce banks to provide credit to riskier firms that would otherwise have to rely exclusively on trade credit.

All else equal, an increase in the supply of credit and a decrease in the demand for credit would lead to lower interest rates. These movements might also be expected to influence the competitive position of banks and the contractual terms of loans, which in turn may influence the bank's desire for risk. Consistent with this conjecture, [Qian and Strahan \(2007\)](#) find that creditor rights influence the structure of the loan contract and lower the average interest rates on bank loans.

Lastly, creditor rights protection may also influence creditors' incentive to monitor their borrowers after they sign the loan contract. In general, asymmetric information and differing incentives may induce borrowers to engage in asset substitution, where they use the funds to invest in riskier projects. A fundamental principle of credit risk management for banks is to write covenants into loan contracts that restrict borrowers from engaging in riskier activities. By monitoring borrowers' activities to see whether they are complying with the covenants and by enforcing the covenants if they are not, lenders can reduce or prevent the moral hazard behavior on the part of borrowers.⁶

⁴ Also, [Vig \(2007\)](#) finds that improvement in the rights of secured creditors in India led to a decrease in usage of secured debt by firms.

⁵ Because of the concentrated holdings and access to information, banks have the ability to exert much greater influence and pressure on management than do public debt holders ([Denis and Mihov, 2003](#)). [Denis and Mihov \(2003\)](#) find that firms with the highest credit quality borrow from public sources, firms with medium credit quality borrow from banks, and firms with the lowest credit quality borrow from non-bank private lenders. They also find (weak) evidence that firm managers seek to avoid the influence, scrutiny, and pressure that banks and other private lenders would exert by issuing public debt.

⁶ According to [Rajan and Winton \(1995\)](#), covenants and collaterals are both contractual devices that increase a lender's incentive to monitor. [Sufi \(2007\)](#) recently found evidence that certain observed differences in the structures of syndicated loans can be attributed to the need to monitor borrowers differing in the amount of asymmetry information.

² While these two channels are certainly not mutually exclusive, [DMS \(2007\)](#) find that creditor rights are relatively more important in more developed countries, whereas information sharing is relatively more important in promoting lending in poorer countries. The same channels that affect the level of private lending are also likely to affect the level of bank risk-taking.

³ For example, [Gilson \(1989\)](#) shows that managers dismissed as a result of financial distress suffer a significant private loss in the form of future employment opportunities. [Eckbo and Thorburn \(2003\)](#) find that managers of bankrupt firms in Sweden suffer a median income loss of 47%.

However, it may be costly for lenders to monitor and enforce the various covenants that arise from more complicated loan contracts.⁷ A lender therefore balances the marginal benefit of monitoring a given borrower with the cost of doing so, when determining the scope/number of covenants and the level of subsequent monitoring. With improved creditor rights (CR) protection, the losses of the creditor in the state of a default decrease. Thus, the marginal benefit of monitoring necessarily declines, implying a lower equilibrium level of monitoring effort by the creditor, and a greater probability of default.⁸ This reduction in monitoring is a response by a rational lender to a favorable change in the legal environment which leads to greater protection to the lender should the borrower default. It should be noted that a larger probability of default does not mean that the lenders' ex post losses are greater. With greater probability of default but greater protection (i.e., smaller losses) in the case of a default, lenders' actual losses may either rise or fall.

While the above discussion focuses on how creditor rights influence contracting and monitoring between banks and their borrowers, creditor rights may also influence the willingness of creditors to provide funds to the bank itself as either depositors or debt holders. In many respects, these effects may be mutually reinforcing, i.e., the same effects that influence a debtholders' willingness to invest in a bank are likely to be the same factors that influence a bank's willingness to engage in risky lending. However, another key variable that comes into play here is whether the bank is operating in a country with deposit insurance. Consequently, in our subsequent analysis, we also control for whether there is deposit insurance and then interact this variable with the measures of creditor rights.

Overall, the strength of creditor rights arguably has a number of complex effects on lender's and borrower's ex ante risk incentives, lender's ex post incentives for monitoring, the likelihood that bankruptcy is declared (see Claessens and Klapper, 2005), and the recovery rates in bankruptcy. In the same regard, information sharing is also likely to influence the risk of any given loan. Information sharing registries (private bureau and/or public registry) are the institutional mechanisms developed by the banks and other creditors to mitigate the information asymmetry problem. Specifically, for a variety of reasons, information sharing among creditors helps reduce both the adverse selection and moral hazard problems associated with bank lending. First, information sharing improves banks' knowledge of applicants' characteristics and permits more accurate predictions of repayment probability (Jappelli and Pagano, 2002). This allows lenders to target and price their loans more accurately, thereby easing adverse selection problems. Pagano and Jappelli (1993) show that information sharing helps mitigate adverse selection concerns by

improving the pool of borrowers. Information sharing also works as a disciplining device for borrowers. In an environment with information sharing, borrowers realize that if they default, their reputation with all other potential lenders is compromised, which in turn will make it more difficult and/or more expensive to receive future credit. Consequently, by enhancing borrowers' incentives to repay, information sharing tends to reduce the moral hazard problems that arise after the contract is signed (Vercammen, 1995; Padilla and Pagano, 1997, 2000). Information sharing is therefore likely to reduce the risk of any given loan.

The effects of creditor rights and information sharing on total bank risk taking are likely to be even more complex. Creditor rights and information sharing not only affect borrower's incentives for risk taking, which influence the risk of any specific loan, but they also may influence a bank's willingness to lend to riskier borrowers, the bank's mix of securities and loans, and the bank's willingness to hold capital—all of which have important effects on the bank's overall level of risk. Moreover, information sharing reduces the adverse selection problem in making loan decisions and the post-lending moral hazard problem, which induce the positive link between creditor rights and bank risk taking. More specifically, banks' increased willingness to lend to riskier borrowers due to increased creditor rights may not be as strong if there is information sharing (especially the sharing of negative information).⁹ In addition, as mentioned above, information sharing itself works as a post-lending disciplining/monitoring device for borrowers. This might also attenuate the impacts of weaker post-lending monitoring caused by the stronger creditor rights. It is therefore likely that information sharing will attenuate the positive effects of creditor rights on bank risk taking. In other words, there might be an interactive effect between creditor rights and information sharing on bank risk taking. Furthermore, their effects on bank risk taking are likely to be heavily influenced by the bank regulatory environment. In this regard, we view the connection between creditor rights, information sharing, and bank risk taking as ultimately an empirical question.

3. Data and summary statistics

3.1. The sample

The data used in this study are compiled from five main sources:

- (1) Bank-level accounting information for nearly 2,400 banks from 69 countries is obtained from the BankScope database provided by Bureau van Dijk and Fitch Ratings. The BankScope database has comprehensive coverage in most countries, accounting for over 90% of all banking assets in each country. Each

⁷ See, e.g., Holmstrom and Tirole (1997) and Allen, Carletti, and Marquez (2008).

⁸ This is confirmed by our regression analysis using the DealScan data that the number of covenants of bank loan contracts, as a proxy of monitoring intensity of banks, declines with the level of creditor rights protection across countries.

⁹ Most information sharing agencies will report negative/black information (e.g., late payments, defaults, and arrears), though only some of them will report both negative and positive information (e.g., credit exposure in good conditions, and patterns of on-time payments) as well (Jappelli and Pagano, 2002; Barth, Lin, Lin, and Song, 2009).

bank report contains a detailed balance sheet and income statement totaling up to 200 data items and 36 pre-calculated financial ratios. In this study, we mainly use the recent data reported over the period 2000–2007;

- (2) The DMS (2007) and World Bank “Doing Business” data sets provide cross-country information regarding creditor rights and information sharing measures during the past 30 years;
- (3) Measures indicating whether a country had a banking crisis in any given year are obtained from the database compiled by Honohan and Laeven (2005). Their banking crisis database covers the systematic banking crises in 93 countries during the sample period of 1972–2002;
- (4) The Barth, Caprio, and Levine (2006) (BCL henceforth) data set on international bank supervision and regulation; the data set was firstly composed in 1999 and was recently updated in 2008;
- (5) The Industrial Statistics Database (INDSTAT4) published in 2008 by the United Nations Industrial Development Organization (UNIDO) on the industrial-level data of share and growth rate in real value added. We use the data on 36 industries covered by the external finance dependence index developed by Rajan and Zingales (1998) in 79 countries over the sample period of 1996–2005.

In addition to the five data sets mentioned above, we rely on three other data sources. The 2007 World Development Indicator (WDI) and the World Governance Indicators compiled by Kaufmann, Kraay, and Mastruzzi (2006) are used to control for macroeconomic and institutional factors that might affect the overall level of bank efficiency in a country. The deposit insurance coverage data across 89 countries over the time period 1960–2004 are obtained from the comprehensive deposit insurance database developed by Demirgüç-Kunt, Kane, and Laeven (2008).

Table 1 presents the summary statistics for all of the key variables. The variable definitions and the data sources are described in Appendix A.

3.2. Bank risk taking

Our primary measure of bank risk taking is the Z-score of each bank, which equals the return on assets plus the capital-asset ratio divided by the standard deviation of asset returns. Specifically, $Z\text{-score} = (ROA + CAR) / \sigma(ROA)$, where ROA is the rate of return on assets, CAR is the ratio of equity to assets, and $\sigma(ROA)$ is an estimate of the standard deviation of the rate of return on assets, all measured with accounting data. Intuitively, the measure represents the number of standard deviations below the mean by which profits would have to fall so as to just deplete equity capital (Boyd, De Nicolò, and Al Jalal, 2006). As a measure of a bank's distance from insolvency (Roy, 1952), Z-score has been widely used in the recent literature (e.g., Laeven and Levine, 2009). A higher value of Z-score indicates more bank stability of the bank. Since

the Z-score is highly skewed, we follow Laeven and Levine (2009) and use the natural logarithm of the Z-score as the risk measure. For brevity, we use the label “Z-score” in referring to the logged Z-score in the remainder of the paper. In our analysis, we collect the data for 2,386 banks across 69 countries over the period from 2000 to 2007 using the BankScope database.¹⁰ The ROA and capital-asset ratio are therefore calculated as the mean over 2000–2007, and $\sigma(ROA)$ is the standard deviation of ROA estimated over the time period 2000–2007.¹¹

In addition, in order to get some sense about which component of the Z-score is principally driving the relationship between the independent variables (e.g., creditor rights, information sharing) and Z-score, we use the three components of Z-score (i.e., ROA, CAR, and $\sigma(ROA)$) as separate dependent variables. As a further robustness test, we also use the volatility of net interest margin as an alternative risk measure—where we measure volatility as the standard deviation of net interest margin over the period 2000–2007.

Looking at the summary statistics in Table 1, we see that the mean log Z-score is 3.240, and that the standard deviation is 1.086. This fairly high standard deviation and the wide range in Z-scores suggest that there is considerable cross-sectional variation in the level of bank risk. These summary statistics are similar to those reported by Laeven and Levine (2009); in looking at a smaller sample (287 banks in 33 countries), they report a mean Z-score of 2.85 and a standard deviation of 0.99.

3.3. Creditor rights and information sharing

As can be seen in Table 1, Panel A, the average creditor rights score across countries and years is 2.018, and the average scores range from zero to four. Looking at the various components of the creditor rights index for the full sample, *Restrictions on reorganization* is found in 30.3% of the observations; *No automatic stay* is found in 42.4% of the observations; *Secured creditor paid first* is found in 68.2% of the observations; and *No management stay* is found in 56.1% of the observations.

Once again, the depth of credit information index ranges from zero to six, with higher values indicating the availability of more credit information, from either a public registry or a private bureau, to facilitate lending decisions. As can be seen from Table 1, Panel A, the mean value of the index is 3.515. Furthermore, in a more detailed analysis, we also use two dummy variables (i.e., *Public registry* and *Private bureau*) to explore their impacts on bank risk taking, respectively. In our sample, there is at least one private bureau in 56% of the countries and one public registry in 47% of the countries. We also use the coverage of public registries (*Public registry coverage*) and private bureaus (*Private bureau coverage*)

¹⁰ The data were trimmed to reduce the influence of outliers. The banks with the upper and lower 1% of the ROA and CAR were excluded from our analysis. The banks with missing observations (e.g., banks with only one or two years of data) were also excluded from the sample.

¹¹ We also tried different time slots (e.g., 2001–2005, 2003–2007) and find the empirical results highly robust.

Table 1

Summary statistics.

<i>Panel A: Bank risk taking analysis</i>						
Variable	Mean	Std. dev.	Min	Max	No. of countries	No. of obs.
<i>Bank-level data</i>						
Log Z-score	3.24	1.086	−4.877	5.296	69	2386
ROA	0.019	0.027	−0.103	0.407	66	2386
CAR	0.089	0.071	0	0.596	64	2386
$\sigma(\text{ROA})$	0.012	0.028	0	0.742	66	2386
Volatility of net interest margin	0.015	0.185	0	3.872	66	2363
Bank growth	0.075	0.24	−0.901	0.998	69	2392
Bank size	12.635	2.414	5.142	21.317	69	2392
Too-big-to-fail	0.062	0.242	0	1	69	2392
Non-performing loans (%)	4.827	10.363	0	85.40	69	2392
Loan loss provisions (%)	2.402	5.218	0	100	69	2392
<i>Country-level data</i>						
Creditor rights	2.018	1.231	0	4	69	69
Restrictions on reorganization (cr1)	0.303	0.463	0	1	66	66
No automatic stay (cr2)	0.424	0.498	0	1	66	66
Secured creditor paid first (cr3)	0.682	0.469	0	1	66	66
No management stay (cr4)	0.561	0.5	0	1	66	66
Information sharing	0.833	0.376	0	1	66	66
Private bureau dummy	0.560	0.500	0	1	66	66
Public registry dummy	0.470	0.503	0	1	66	66
Depth of credit information	3.515	1.94	0	6	66	66
Private bureau coverage	27.082	32.785	0	100	66	66
Public registry coverage	4.58	11.523	0	65.18	66	66
Min. capital-asset ratio requirement	0.091	0.015	0.08	0.12	66	66
GDP volatility	1.794	1.809	0.436	4.095	66	66
Control of corruption	−0.033	1.033	−1.244	2.238	66	66
Government effectiveness	0.151	0.919	−1.421	2.139	66	66
Political stability	−0.461	0.831	−2.491	1.205	66	66
Quality of regulation	0.195	0.842	−2.2	1.887	66	66
Rule of law	−0.072	1.021	−1.471	1.986	66	66
Voice and accountability	−0.068	0.914	−1.654	1.505	66	66
Transparency	0.545	0.778	−1.21	2.03	66	66
Diversification index	1.394	0.595	0	2	69	69
Overall capital stringency	4.409	1.285	1	7	69	69
Financial statement transparency	4.881	0.684	3	6	69	69
Deposit insurance coverage	1.122	1.541	0.172	7.579	69	69
HHI	0.194	0.113	0.021	0.733	69	69
Log GDP per capita	9.353	1.194	5.283	11.178	69	69
Log GDP	24.279	1.822	21.361	30.132	69	69
Inflation (%)	18.352	78.938	−14.17	368.883	69	69
Note: The 69 countries include Algeria, Australia, Austria, Azerbaijan, Belgium, Botswana, Brazil, Bulgaria, Canada, Chile, Colombia, Costa Rica, Cote d'Ivoire, Croatia, Czech Republic, Denmark, Egypt, El Salvador, France, Germany, Greece, Guatemala, Hong Kong, India, Ireland, Italy, Japan, Jordan, Kazakhstan, Kenya, Rep. Korea, Kyrgyz Republic, Lebanon, Lithuania, Macedonia (FYR), Mexico, Moldova, Morocco, Namibia, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, Pakistan, Panama, Papua New Guinea, Philippines, Poland, Portugal, Romania, Russian Federation, Rwanda, Senegal, Singapore, Slovak Republic, Slovenia, South Africa, Spain, Sri Lanka, Switzerland, Togo, Tunisia, Ukraine, United Kingdom, United States, Uruguay, Venezuela, and Zimbabwe.						
<i>Panel B: Banking crisis analysis</i>						
Variable	Mean	Std. dev.	Min	Max	No. of countries	No. of obs.
Systemic banking crisis dummy	0.162	0.369	0	1	93	2516
Creditor rights	1.82	1.203	0	4	93	2516
Information sharing	0.655	0.475	0	1	93	2516
Private bureau dummy	0.362	0.481	0	1	93	2516
Public registry dummy	0.391	0.488	0	1	93	2516
Log GDP per capita	7.504	1.661	4.533	10.545	93	2516
Log GDP	23.938	2.159	19.806	29.917	93	2516
Terms of trade change (%)	10.23	44.791	−23.813	315.625	93	2516
Real interest rate (%)	7.483	48.830	−125.17	374.308	93	2516
Inflation (%)	49.132	477.937	−29.173	6836.88	93	2516
M2/reserves	11.6	195.3	0.114	2492.88	93	2516
Depreciation of domestic currency (%)	1.566	54.567	−0.999	57.037	93	2516
Credit growth (%)	1.294	3.978	−0.294	33.318	93	2516
Bank concentration	0.831	0.191	0.229	1	93	2516
Deposit insurance coverage	0.841	1.772	0	7.579	93	2516

Table 1. (continued)

Note: All the variables are time varying. The creditor rights and information sharing measures are from [DMS \(2007\)](#). The time period coverage is from 1980 to 2007. In the regression analysis, the independent variables are lagged by two years. The 93 countries include Algeria, Angola, Argentina, Australia, Austria, Bangladesh, Belgium, Benin, Bolivia, Botswana, Brazil, Burkina Faso, Burundi, Cameroon, Canada, Chad, Chile, China, Colombia, Rep. Congo, Costa Rica, Cote d'Ivoire, Denmark, Dominican Republic, Ecuador, Egypt, El Salvador, Ethiopia, Finland, France, Germany, Ghana, Greece, Guatemala, Haiti, Honduras, Hong Kong, Hungary, India, Indonesia, Iran, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kenya, Rep. Korea, Lesotho, Madagascar, Malawi, Malaysia, Mali, Mauritania, Mexico, Morocco, Mozambique, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, Norway, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Rwanda, Senegal, Sierra Leone, Singapore, South Africa, Sri Lanka, Sweden, Switzerland, Tanzania, Thailand, Togo, Tunisia, Turkey, Uganda, United Kingdom, United States, Uruguay, Venezuela, Zambia, and Zimbabwe.

Panel C: Industry growth analysis

Variable	Mean	Std. dev.	Min	Max	No. of countries	No. of obs.
Growth rate in real value added for 1996–2005	0.01	0.204	−0.989	0.991	79	2430
Share in value added in 1996	0.027	0.046	0	0.956	79	2430
Creditor rights	1.843	1.241	0	4	79	79
Information sharing	0.848	0.361	0	1	79	79
Deposit insurance coverage	1.084	1.681	0	7.579	79	79
Depth of credit information	3.615	1.972	0	6	79	79
HHI	0.25	0.155	0.037	0.939	79	79
Stock market capitalization in 1996	0.345	0.516	0	2.635	79	79
Private credit in 1996	0.421	0.392	0.012	2.083	79	79

Note: The 79 countries include Albania, Argentina, Australia, Austria, Azerbaijan, Belgium, Bolivia, Brazil, Bulgaria, Cameroon, Canada, Chile, Colombia, Costa Rica, Cote d'Ivoire, Czech Republic, Denmark, Ecuador, Egypt, El Salvador, Ethiopia, Finland, France, Georgia, Germany, Greece, Hong Kong, Hungary, India, Indonesia, Iran, Ireland, Israel, Italy, Japan, Jordan, Kenya, Kuwait, Kyrgyz Republic, Latvia, Lithuania, Macedonia (FYR), Madagascar, Malawi, Malaysia, Mexico, Moldova, Mongolia, Morocco, Nepal, Netherlands, Norway, Oman, Panama, Paraguay, Peru, Philippines, Poland, Portugal, Rep. Korea, Rep. Yemen, Romania, Russian Federation, Senegal, Singapore, Slovak Republic, South Africa, Spain, Sri Lanka, Sweden, Switzerland, Tanzania, Thailand, Tunisia, Turkey, Uganda, United Kingdom, Uruguay, and Vietnam.

to proxy for the information content of the information agencies. Specifically, the *Public registry coverage* indicator reports the number of individuals and firms listed in a public credit registry with information on repayment history, unpaid debts, or credit outstanding from the past five years, scaled by the adult population. The *Private bureau coverage* indicator reports a similar coverage ratio for private bureaus.

3.4. Measuring banking crises

We also explore the effects of creditor rights and information sharing on the likelihood of financial crisis within a country's banking system. *Banking crisis* is a dummy variable that takes on the value one if the country is going through a systemic crisis in that year, and zero if it is not. A systemic crisis is a situation in which significant segments of the banking sector become insolvent or illiquid, and cannot continue to operate without special assistance (such as bank holidays, deposit freezes, blanked guarantees to depositors or other bank creditors, or large scale nationalizations) from the monetary or supervisory authorities ([Demirgüç-Kunt and Detragiache, 2002](#); [Beck, Demirgüç-Kunt, and Levine, 2006](#)). The banking crisis data are obtained from the banking crisis database, a comprehensive database of banking crisis episodes, compiled by [Honohan and Laeven \(2005\)](#). It is a panel data set with a sample period of 1972–2007 covering 93 countries. Looking at the summary statistics in Table 1 (Panel B), we see that banking crises are fairly common: in our sample of 93 countries over a

28-year sample period, we see that the likelihood of crisis is 16.2%.

As [Boyd, De Nicolo, and Loukoianova \(2009\)](#) point out, most of the existing banking crisis indexes (e.g., [Honohan and Laeven, 2005](#); [Reinhart and Rogoff, 2008](#)) depend on information obtained from bank regulators and central banks, and they define the beginning of the crisis as the time in which the government first responded to the perceived crisis. However, government responses to banking distress may occur with a lag because of asymmetric information, the extent of banking distress, and various political considerations. Therefore, equating the dating of a government response to banking distress to the dating of a systemic bank crisis may systematically lag the actual arrival of systemic shocks ([Boyd, De Nicolo, and Loukoianova, 2009](#)). Furthermore, this systematic lag might make it difficult for researchers to disentangle the effects of a negative shock to the banking industry from those of the positive restorative policy response ([Boyd, De Nicolo, and Loukoianova, 2009](#)). To address this concern, we follow [Boyd, De Nicolo, and Loukoianova \(2009\)](#) by using another set of theory-based indicators of systemic bank shocks (SBS) to test the robustness of the results. Specifically, we create two dummy variables, SBSL10 and SBSL25, that are set equal to one if real domestic lending growth is lower than the 10th and 25th percentile of the entire distribution of real domestic bank credit growth across countries, respectively. We also use SBSL10 and SBSL25 as two bank crisis indicators, which equal one if the growth rate of the deposit-to-gross-domestic-product (GDP) ratio is lower than the 10th and 25th percentile of its distribution across countries respectively.

3.5. Sector growth and external finance dependence

In our analysis, we also investigate the impacts of creditor rights and credit information sharing on real industrial growth. Following [Rajan and Zingales \(1998\)](#), we measure industrial growth as the real growth in industry value added (annually compounded). The data on value added for each industry in each country are obtained from the Industrial Statistics Database (INDSTAT4), which was published by UNIDO in 2008. We calculate the growth rate in real value added over the period 1996–2005 for each industry in each country. The industry-level data on external finance dependence (EFD) are from [Rajan and Zingales \(1998\)](#), who construct their index at the industry level for a sample of US firms.¹² They argue that for technological reasons, such as scale economies, gestation period, the cash harvest period, and intermediate product intensity, some industries might rely more heavily on external finance than others. They further argue that the financial dependence of US industries can serve as an appropriate benchmark because the relatively frictionless, sophisticated, and developed US financial markets should allow US firms to encounter fewer obstacles to achieving their desired financial structure than firms in other countries. This approach thus provides a valid and exogenous way to evaluate the extent of an industry's external dependence anywhere in the world ([Kroszner, Laeven, and Klingebiel, 2007](#)). We expect that the impacts of creditor rights and information sharing on sector growth would be stronger in the industries with more external financial dependence. We calculate the real growth in value added for 36 sectors defined on the mixture of three- and four-digit industries International Standard Industrial Classification (ISIC) codes used by [Rajan and Zingales \(1998\)](#). The final sample used in our growth regression analysis includes 2,430 industry observations in 79 countries. The mean external financial dependence ratio of the industries in our sample is 31.9%, and the mean growth rate in value added is 1%.¹³

3.6. Additional bank regulation and bank controls

As measures of banking regulation, we also control for *Capital stringency*, *Diversification Index*, *Financial statement transparency* and *Deposit insurance coverage*. The first three variables are from the banking regulation database compiled by [Barth, Caprio, and Levine \(2006\)](#). The database is based on a World Bank survey on bank regulation and supervision in 152 countries over the period 1999–2007. The *Deposit insurance coverage* is obtained from [Demirgüç-Kunt, Kane, and Laeven \(2008\)](#).

Capital stringency is constructed from several variables that indicate whether the capital requirement reflects certain risk elements. This measure takes into account whether the minimum capital-asset ratio requirement is in line with the Basel Committee on Banking Supervision guidelines; whether the minimum ratio varies as a function of an individual bank's credit risk and market risk; and whether the market value of loan losses not realized in accounting books, unrealized losses in securities portfolios, and/or unrealized foreign exchange losses, are deducted from the book value of capital. *Diversification index* measures whether there are explicit, verifiable, and quantifiable guidelines for asset diversification (e.g., if banks are required to have some minimum diversification of loans among sectors, or if there are sectoral concentration limits) and whether banks are allowed to make loans abroad. Higher values of the index indicate more bank asset diversification.

Financial statement transparency measures the transparency of bank financial statement practices. It includes the information on whether accrued, though unpaid, interest/principal enter the income statement; whether financial institutions are required to produce consolidated accounts covering all bank and any non-bank financial subsidiaries; whether off-balance sheet items are disclosed to the public; whether banks are required to disclose their risk management procedures to the public; and whether bank directors are legally liable if information disclosed is erroneous or misleading. Higher values indicate better financial statement transparency. *Deposit insurance coverage* is the ratio of deposit insurance coverage to deposits per capita. [BCL \(2006\)](#) point out that deposit insurance intensifies the moral hazard problem in banking because depositors no longer face the risk of losing their savings, which diminishes their incentives to monitor bank activities. [Laeven \(2002\)](#) finds a positive link between costs of deposit insurance and bank distress.

Regarding the banking market structure, we use the standard Herfindahl–Hirschman Index (HHI), which is equal to the sum of the squares of the market shares (deposits) of each individual bank in individual countries. The (normalized) Herfindahl–Hirschman Index ranges from zero to one with a higher value indicating greater monopoly power. The HHI index is calculated using the BankScope database. We also use the market share of the three largest banks in total bank assets (*Banking concentration*) to measure banking concentration. In the traditional literature, higher concentration is generally linked to a lower level of competition within the banking industry. We also control for several bank characteristics. *Bank size* equals the logarithm of total bank assets in millions of US dollars. The average *Bank size* over the period 2000–2007 is calculated and used in the regressions. We include a square term of *Bank size* into the regressions to capture the potential non-linear impacts of bank size on bank risk taking. *Bank growth* is the average growth rate of bank revenue over the period 2000–2007. *Too-big-to-fail* is a dummy variable that takes a value of one if the bank's share in the country's total deposits exceeds 10%.

¹² External financial dependence measures the fraction of desired investment that cannot be financed with internal cash flows.

¹³ Our 1% mean growth rate in real value added is somewhat lower than 3.3% reported in [Rajan and Zingales \(1998\)](#) and [Beck and Levine \(2002\)](#). This is due to the fact that our new UNIDO 2008 sample covers 79 countries over the period of 1996–2005. The Industrial Statistics Yearbook 1993 database used by [Rajan and Zingales \(1998\)](#) and [Beck and Levine \(2002\)](#) covers 41 countries over 1980–1990.

3.7. Country controls

We also include several country-level variables to control for differences in economic development and institutions across countries. First, we include GDP per capita to capture the economic development of the region/country. Second, we include the natural logarithm of GDP to capture the size of the economy. We also control for the inflation of the economy. Furthermore, we include a series of other political and institutional quality indexes as a check on the robustness of the results. The World Governance Indicators (Kaufmann, Kraay, and Mastruzzi, 2006) are constructed from 276 individual variables taken from 31 different sources produced by 25 different organizations. The indexes measure different dimensions of governance, which can be summarized as follows:

- (1) *Government effectiveness (Government effective)*—the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies.
- (2) *Political stability and absence of violence (Political stability)*—perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means, including political violence and terrorism.
- (3) *Regulatory quality (Regulation)*—the ability of the government to formulate and implement sound policies and regulations that permit and promote market competition and private-sector development.
- (4) *Rule of law (Law)*—the extent to which agents have confidence in and abide by the rules of society, and in particular, the quality of contract enforcement, the police, and the courts, as well as the likelihood of crime and violence.
- (5) *Voice and accountability (Voice)*—the extent to which a country's citizens are able to participate in selecting their government, as well as the extent to which they enjoy freedom of expression, freedom of association, and a free media.
- (6) *Control of corruption (Control of corruption)*—the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as “capture” of the state by elites and private interests. Higher values indicate better control of corruption.

In addition, we also control for the economic and political transparency of the country using the *Transparency Index* developed by Kaufmann and Bellver (2005). A detailed explanation of the index can be found in Appendix A. Furthermore, in the analysis of banking crisis, we follow Demirgüç-Kunt and Detragiache (2002) to control for some additional macroeconomic factors such as the terms of trade, M2/reserve ratio, credit growth, and the depreciation of domestic currency. Detailed definition, of these variables can also be found in Appendix A.

4. Empirical results regarding bank risk taking

4.1. Does the level of creditor rights and information sharing influence bank risk taking?

First, we examine the impact that creditor rights and information sharing have on bank risk taking. Our main dependent variable is the log Z-score, and the key independent variables are the creditor rights index, and two measures of information sharing—a dummy variable indicating whether an information sharing agency operates in the country, and the index measuring the depth of available credit information. Specifically, the regression analysis is expressed as follows:

$$Z_{ij} = \alpha + \beta'_1 \text{ Creditor Rights Measures}_j + \beta'_2 \text{ Information Sharing Measures}_j + \beta'_3 \text{ Bank Regulation Controls}_j + \beta'_4 \text{ Bank Controls}_{ij} + \beta'_5 \text{ Macro Controls}_j + \varepsilon_{ij}, \quad (1)$$

where the i and j subscripts indicate bank and country, respectively. Z_{ij} is log Z-score, α is a constant, and β'_k is a vector of parameters. The creditor rights measure is based on the aggregate creditor right index (*Creditor rights*), which has been defined in Section 2, and is calculated as the average measure over the sample time period. Information sharing measures include the *Information sharing* dummy variable, and *Depth of credit information*, both of which have been defined in Section 2. Bank regulation controls include *Diversification index*, *Capital stringency*, *Financial statement transparency*, *Deposit insurance coverage*, and banking market *HHI* index. Bank controls include *Bank size*, *Bank size squared*, *Too-big-to-fail*, and *Bank growth*. The detailed definitions of these measures can be found in Section 3 and Appendix A. Macro variables include log GDP per capita, log GDP, and inflation. In addition, the heteroskedasticity-robust standard errors clustered at the country level are used in computing t -values. When clustering the standard errors by country, observations are not restricted to be independent within countries; rather, observations are required to be independent across countries. This treatment has been shown to be important in deriving robust and reliable results (Beck, Demirgüç-Kunt, and Peria, 2007). Our regression results are reported in Table 2.

For each of the reported specifications, we consistently find that a higher creditor rights index translates into higher levels of bank risk taking. (Once again, a higher estimated Z-score implies more stability, i.e., less risk taking). As can be seen from the table, the coefficients of creditor rights are negative and statistically significant, suggesting that the net effect of creditor rights on bank risk taking is positive and significant. A one standard deviation increase in *Creditor rights* (1.203) is associated with a change in Z-score of about -0.30 ($= -0.25 \times 1.203$), where the mean of Z-score is 3.24. We also find that higher levels of information sharing are associated with lower levels of bank risk taking. This result holds for each of the information sharing proxies (*Information sharing* and *Depth of credit information*). Specifically, the presence of an information sharing agency will increase the Z-score

Table 2

Z-score and creditor rights: Bank level basic OLS regressions.

The dependent variable is the logarithm of Z-score. $Z\text{-score} = (\text{ROA} + \text{CAR}) / \sigma(\text{ROA})$, where $\text{ROA} = \pi/A$ is return on assets and $\text{CAR} = E/A$ is capital-asset ratio, both averaged over 2000–2007. $\sigma(\text{ROA})$ is the standard deviation of ROA over 2000–2007. Higher Z-score implies more stability. The estimation is based on OLS regressions. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3	4	5
Creditor rights	−0.232 [0.023]**	−0.244 [0.006]***	−0.266 [0.012]**	−0.254 [0.033]**	−0.257 [0.035]**
Information sharing		0.576 [0.013]**	0.614 [0.003]***		
Information sharing × Creditor rights			0.274 [0.035]**		
Depth of credit information				0.601 [0.005]***	0.598 [0.021]**
Depth of credit information × Creditor rights					0.111 [0.037]**
Diversification index	0.328 [0.037]**	0.394 [0.011]**	0.367 [0.014]**	0.306 [0.054]*	0.421 [0.028]**
Overall capital stringency	0.141 [0.054]*	0.176 [0.041]**	0.163 [0.047]**	0.120 [0.058]*	0.196 [0.044]**
Financial statement transparency	0.272 [0.030]**	0.238 [0.059]*	0.285 [0.028]**	0.243 [0.022]**	0.245 [0.027]**
Deposit insurance coverage	−0.137 [0.016]**	−0.140 [0.010]**	−0.138 [0.023]**	−0.153 [0.000]***	−0.151 [0.013]**
Bank size	0.398 [0.019]**	0.412 [0.007]***	0.423 [0.004]***	0.437 [0.008]***	0.436 [0.004]***
Bank size squared	−0.013 [0.018]**	−0.015 [0.012]**	−0.014 [0.021]**	−0.016 [0.024]**	−0.014 [0.026]**
Too-big-to-fail	−0.273 [0.027]**	−0.281 [0.024]**	−0.308 [0.016]**	−0.285 [0.023]**	−0.287 [0.022]**
HHI	−0.101 [0.073]*	−0.094 [0.119]	−0.122 [0.080]*	−0.110 [0.079]*	−0.113 [0.071]*
Bank growth	−0.043 [0.141]	−0.067 [0.079]*	−0.057 [0.112]	−0.068 [0.082]*	−0.059 [0.106]
Log GDP per capita	0.186 [0.053]*	0.215 [0.042]**	0.252 [0.037]**	0.261 [0.039]**	0.278 [0.014]**
Log GDP	0.145 [0.025]**	0.131 [0.037]**	0.127 [0.046]**	0.102 [0.131]	0.113 [0.064]*
Inflation	−0.133 [0.000]***	−0.128 [0.008]***	−0.145 [0.005]***	−0.108 [0.000]***	−0.107 [0.002]***
Constant	1.098 [0.036]**	1.182 [0.035]**	1.287 [0.033]**	1.181 [0.026]**	1.181 [0.017]**
Observations	2386	2363	2363	2363	2363
Adjusted R^2	0.251	0.252	0.254	0.261	0.261
Countries	69	66	66	66	66

by 0.576; a one standard deviation increase in *Depth of credit information* is associated with a change in Z-score of 1.11. Given that the mean of Z-score is 3.24, the effects are not only statistically significant but also economically significant.

As explained in Section 2, stronger creditor rights might induce banks to lend to a wider set of (riskier) borrowers. At the same time, information sharing helps reduce adverse selection problems in loan screening and mitigate the post-lending moral hazard problems. We therefore interact the information sharing proxies with the creditor rights measure to test the potential interactive effects. The empirical results are presented in columns 3 and 5 of Table 2. While the main results continue to hold, we also find that the presence of information sharing tends to mitigate the effects of stronger creditor rights on bank risk taking. In other words, bank risk taking is less sensitive to creditor rights

in the presence of good information sharing. In fact, in the presence of an information sharing agency (column 3), the negative effects of creditor rights on Z-score are mainly eliminated (marginal effect on creditor rights is $-0.266 + 0.274 \times \text{information sharing}$).

Examining the coefficients on the various control variables, we find a number of additional interesting results. As might be expected, the bank regulatory variables have a significant effect on bank risk taking. Consistent with previous studies (e.g., Demirgüç-Kunt and Detragiache, 2002), we find that deposit insurance tends to lead to higher bank risk taking. As Barth, Caprio, and Levine (2006) point out, deposit insurance intensifies the moral hazard problem in banking because depositors no longer face the risk of losing their savings, which diminishes their incentives and efforts at monitoring bank activities. We find an inverse U-shape relationship between bank size and bank risk taking. In addition, banks

with higher growth rates and those classified as too-big-to-fail engage in more risk taking. In contrast, more stringent capital requirements, greater financial statement transparency, and diversification are all correlated with lower risk taking.

We test the robustness of these main results in two ways. First, we obtained additional data from BankScope's historical files. Our expanded data extend over the 12-year period, 1996–2007, which enables us to estimate the results over four three-year non-overlapping subperiods.¹⁴ This extension expands the sample size from 2,400 to nearly 6,800 observations, and allows us to test the robustness of these results in a panel setting, where in all estimations time and country fixed effects are included. Second, we added a series of macro institutional indexes in our model to test the robustness of the results. Specifically, we included the country's minimum capital requirement, GDP volatility, and the six components of the World Governance Indicators (Kaufmann, Kraay, and Mastruzzi, 2006), which capture different aspects of the institutional environment (control of corruption, political stability, government effectiveness, quality of regulation, voice and accountability, and rule of law). The detailed definition of the indexes can be found in Section 3. Because some indexes are highly correlated with each other, we include the indexes individually in the models. The empirical results related to these two extensions are presented in Appendices B and C, respectively. In both cases, the empirical results are highly consistent with our previous findings. Next, we consider some additional robustness tests that go beyond our main results.

4.2. Robustness tests: alternative measures of bank risk taking and disaggregation of the Z-score

Decomposing the Z-score measure, we know that all else equal, higher levels of ROA and higher bank capital ratios translate into higher Z-scores, while a larger standard deviation of ROA translates into lower Z-scores. So, when we find, for example, that stronger creditor rights translate into lower Z-scores, this may be attributed to lower ROA, lower capital ratios, and/or a higher standard deviation. Therefore, it is possible that stronger creditor rights may not necessarily increase the risk of banks assets, but rather the drop in Z-score may instead be attributed to a decline in the average bank capital ratio. To further explore how the various components of the Z-score move in response to stronger creditor rights and higher levels of information sharing, we run regressions treating each of these Z-score components as a separate independent variable. In the regressions with capital ratio as dependent variables, we include the country's minimum capital requirement as an additional control. The data are from the bank regulation data set compiled by Barth, Caprio, and Levine (2006). The empirical results are reported in Table 3.

We see that stronger creditor rights protection is significantly associated with lower levels of ROA, higher

standard deviations of ROA, and higher capital ratios. These results suggest that the lower Z-score results primarily from banks investing in riskier, poorer-performing assets, and it is not driven solely by a reduction in bank capital. Indeed, stronger creditor rights are associated with an increase in bank capital. A potential explanation for this finding is that banks tend to extend more credit in the presence of stronger creditor rights and therefore, need to attract more deposits by signaling its monitoring incentives to the depositors.¹⁵ One way to guarantee monitoring is through the equity capital holding since capital forces banks to internalize the costs of their default, thus ameliorating the limited liability problem banks face due to their extensive reliance on deposit-based financing (Allen, Carletti, and Marquez, 2008). We thus observe a positive link between creditor rights protection and bank capital ratio.

We also see in Table 3 that the measures of information sharing are positively related to ROA, and negatively related to the standard deviation in ROA. Both of these factors work simultaneously to increase the Z-score. We do not find a significant relationship between information sharing and capital ratio. Overall, it follows that greater information sharing improves the quality of assets and leads to lower bank risk. Furthermore, the interactive term between creditor rights and information sharing exerts a positive effect on ROA and a negative effect on volatility, both of which are consistent with our finding using Z-score as a dependent variable. We do not find a significant impact of the interactive term on the bank capital ratio.

As robustness tests, we also consider three other dependent variables as alternative measures of bank risk. First, we consider the raw (no log) Z-score as a dependent variable; this enables us to also include the 33 observations that were dropped from the sample because they had a negative Z-score. We also use the volatility of the bank's net interest margin, and loan loss provisions as a percentage of total loans as alternative measures of bank risk.¹⁶ In each case, the results are qualitatively similar. Stronger creditor rights lead to higher volatility, while greater levels of information sharing reduce volatility. Moreover, the interactive term works in a similar fashion—information sharing helps mitigate the positive effects that stronger creditor rights have on risk taking.

The control variables also yield some interesting and consistent findings. *Diversification* and *Financial statement transparency* are positively associated with ROA and CAR but negatively associated with performance volatility. In contrast, *Capital stringency* is negatively associated with ROA and the volatility of ROA, but positively associated with CAR. The aggregate effect of capital stringency on Z-score is positive. It is interesting (and perhaps some-

¹⁵ As Allen, Carletti, and Marquez (2008) point out, given that monitoring is costly and banks have limited liability, banks are subject to a moral hazard problem in the choice of monitoring effort.

¹⁶ We also ran regressions where the percentage of non-performing loans was used as an alternative measure of risk. These (unreported) results were qualitatively similar to the other results reported in Table 3.

¹⁴ The BankScope regular files only contain data over the past eight years.

Table 3

Cross-section OLS regressions: ROA and CAR and other alternative risk-taking measures.

The dependent variables are ROA in columns 1 and 2 and are CAR for columns 3 and 4, respectively. Following [Boyd, De Nicolò, and Al Jalal \(2006\)](#), ROA is return on assets and CAR is capital-asset ratio, both are averaged over 2000–2007. Higher values of ROA and CAR imply more stability. The dependent variables are standard deviation of ROA for columns 5 and 6, volatility of net interest margin for columns 7 and 8, and raw Z-score (no log) for columns 9 and 10, respectively, over 2000–2007. The raw Z-score is divided by ten in regressions. The estimation is based on OLS regressions. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3	4	5	6	7	8	9	10	11	12
	ROA		CAR		$\sigma(\text{ROA})$		Volatility of net interest margin		Raw Z-score (no log)		Loan loss provision (%)	
Creditor rights	−0.0048 [0.028]**	−0.0049 [0.004]***	0.0023 [0.016]**	0.0023 [0.063]*	0.0173 [0.015]**	0.0171 [0.008]***	0.0185 [0.026]**	0.0186 [0.038]**	−0.480 [0.014]**	−0.504 [0.022]**	0.631 [0.025]**	0.705 [0.026]**
Information sharing	0.0128 [0.014]**		0.0232 [0.153]		−0.0406 [0.022]**		−0.0153 [0.047]**		0.804 [0.002]***		−1.393 [0.017]**	
Information sharing × Creditor rights	0.0035 [0.024]**		0.0051 [0.115]		−0.0243 [0.011]**		−0.0157 [0.025]**		0.510 [0.039]**		−0.179 [0.028]**	
Depth of credit information		0.0063 [0.008]***		0.0070 [0.063]*		−0.0352 [0.008]***		−0.0389 [0.034]**		0.856 [0.014]**		−0.979 [0.031]**
Depth of credit information × Creditor rights		0.0015 [0.023]**		0.0016 [0.171]		−0.0017 [0.027]**		−0.0014 [0.017]**		0.199 [0.029]**		−0.389 [0.014]**
Diversification index	0.0059 [0.032]**	0.0061 [0.067]*	0.0070 [0.053]*	0.0056 [0.043]**	−0.0038 [0.031]**	−0.0051 [0.027]**	−0.0046 [0.013]**	−0.0050 [0.011]**	0.844 [0.020]**	0.727 [0.031]**	−0.150 [0.042]**	−0.181 [0.046]**
Overall capital stringency	−0.0031 [0.082]*	−0.0037 [0.031]**	0.0057 [0.038]**	0.0057 [0.061]*	−0.0037 [0.054]*	−0.0044 [0.011]**	−0.0018 [0.154]	−0.0011 [0.041]**	0.357 [0.063]*	0.365 [0.054]*	−0.210 [0.067]*	−0.290 [0.036]**
Financial statement transparency	0.0066 [0.018]**	0.0060 [0.027]**	0.0155 [0.015]**	0.0146 [0.014]**	−0.0082 [0.014]**	−0.0083 [0.012]**	−0.0153 [0.024]**	−0.0197 [0.166]	0.334 [0.041]**	0.505 [0.042]**	−0.791 [0.059]*	−0.924 [0.013]**
Deposit insurance coverage	−0.0003 [0.026]**	−0.0004 [0.017]**	0.0003 [0.017]**	0.0002 [0.028]**	0.0004 [0.034]**	0.0003 [0.043]**	0.0014 [0.079]**	0.0016 [0.032]**	−0.265 [0.026]**	−0.354 [0.015]**	0.017 [0.026]**	0.013 [0.070]*
Min. capital-asset ratio requirement			0.0062 [0.064]*	0.0054 [0.042]**								
Bank size	0.0098 [0.033]**	0.0104 [0.030]**	0.0164 [0.000]***	0.0166 [0.000]***	−0.0135 [0.040]**	−0.0137 [0.027]**	−0.0184 [0.026]**	−0.0184 [0.022]**	0.606 [0.005]***	0.685 [0.005]***	−0.853 [0.039]**	−0.855 [0.027]**
Bank size squared	−0.0004 [0.034]**	−0.0005 [0.032]**	−0.0002 [0.000]***	−0.0002 [0.000]***	0.0004 [0.041]**	0.0004 [0.045]**	0.0028 [0.017]**	0.0027 [0.019]**	−0.038 [0.023]**	−0.038 [0.030]**	0.022 [0.039]**	0.022 [0.034]**
Too-big-to-fail	0.0015 [0.080]*	0.0011 [0.175]	−0.0013 [0.019]**	−0.0026 [0.037]**	0.0194 [0.009]***	0.0175 [0.009]***	0.0463 [0.036]**	0.0414 [0.044]**	−0.568 [0.021]**	−0.494 [0.027]**	0.496 [0.017]**	0.558 [0.035]**
HHI	0.0124 [0.166]	0.0184 [0.081]*	−0.0451 [0.063]*	−0.0417 [0.071]*	0.0306 [0.097]*	0.0317 [0.117]	0.0228 [0.062]*	0.0197 [0.074]*	−0.362 [0.066]*	−0.336 [0.073]*	0.886 [0.053]*	0.784 [0.062]*
Bank growth	0.0006 [0.036]**	0.0006 [0.043]**	0.0002 [0.148]	0.0003 [0.134]	0.0243 [0.078]*	0.0333 [0.037]**	0.0508 [0.011]**	0.0520 [0.017]**	−0.161 [0.141]	−0.198 [0.123]	−1.576 [0.022]**	−1.622 [0.029]**
Log GDP per capita	0.0024 [0.052]*	0.0035 [0.014]**	0.0044 [0.012]**	0.0042 [0.024]**	−0.0033 [0.030]**	−0.0061 [0.053]*	−0.0020 [0.012]**	−0.0012 [0.140]	0.736 [0.042]**	0.717 [0.015]**	−0.587 [0.015]**	−0.587 [0.029]**
Log GDP	0.0030 [0.122]	0.0037 [0.026]**	0.0025 [0.073]*	0.0023 [0.032]**	−0.0040 [0.022]**	−0.0039 [0.053]*	−0.0088 [0.150]	−0.0113 [0.081]*	0.527 [0.125]	0.449 [0.037]**	−0.064 [0.055]*	−0.084 [0.220]
Inflation	−0.0002 [0.108]	−0.0001 [0.077]*	−0.0003 [0.018]**	−0.0002 [0.090]*	0.0003 [0.051]*	0.0004 [0.034]**	0.0002 [0.021]**	0.0001 [0.013]**	−0.362 [0.006]***	−0.224 [0.002]***	0.030 [0.071]*	0.025 [0.011]**
Constant	−0.2188 [0.015]**	−0.3041 [0.021]**	−0.2553 [0.013]**	−0.2221 [0.021]**	0.3582 [0.006]***	0.3459 [0.009]***	0.1630 [0.031]**	0.1447 [0.032]**	3.824 [0.040]**	2.647 [0.019]**	2.857 [0.066]*	2.704 [0.023]**
Observations	2,363	2,363	2,321	2,321	2,363	2,363	2,363	2,363	2,396	2,396	2,363	2,363
Countries	66	66	64	64	66	66	66	66	66	66	66	66
Adjusted R^2	0.187	0.183	0.198	0.224	0.163	0.162	0.2650	0.2510	0.188	0.198	0.142	0.127

what surprising) to note that the results in Table 3 indicate that stronger creditor rights lead to both higher risk taking and lower ROA. While the ROA results are partially mitigated if there is information sharing, it is still somewhat puzzling to find that stronger creditor rights are correlated with lower risk-adjusted returns for banks.

As we discussed in Section 2, stronger creditor rights are likely to have a multitude of influences on a bank's willingness to lend to a given borrower, the contractual terms of loans, and the relative supply and demand of credit. Stronger creditor rights may induce banks to provide loans to a broader set of (riskier) borrowers. If so, we might expect that banks would be willing to provide such loans only if they lead to higher risk-adjusted expected returns. However, at the same time, while stronger creditor rights are likely to lead to an increase in a bank's willingness to supply loans to any given borrower, they are also likely to lead to a reduction in borrowers' demand for loans. The shifting demand for loans is particularly likely if (as Acharya, Amihud, and Litov, 2009 suggest) stronger creditor rights induce firms to reduce leverage and seek other sources of funding. The expected net effect of these shifts in the supply and demand curves is a reduction in equilibrium loan rates. With this issue in mind, we ran (unreported) regressions using DealScan data on about 15,000 syndicate loan contracts of about 3,600 firms in more than 40 countries over the period 1996–2007, where we used the loan spread (measured as the all-in spread measured in basis points over London interbank offered rate (LIBOR)) as the dependent variable. The results confirmed that stronger creditor rights are correlated with lower spreads. These findings also support the findings of Qian and Strahan (2007) who find a negative link between creditor rights and loan rates.

The shifts in the supply and demand curve for loans may also work to reduce the bank's relative bargaining position, in part because the shifting supply curve may coincide with increased competition among banks and other creditors and in part, because stronger creditor rights induce borrowers to shift away from bank loans to other sources of financing. If so, these factors could potentially lead to a reduction in banks' risk-adjusted returns.¹⁷ From a somewhat different perspective, these findings could also be explained in the context of some recent findings by Maddaloni and Peydró (2009). They show that banks tend to soften their credit standards (which could lead to lower risk-adjusted returns) in markets with a lower sustained lending rate.

¹⁷ To further examine the impact that creditor rights have on the contractual terms, we also ran (unreported) regressions using DealScan data where the dependent variable measured the number of covenants in the loan agreement. We find that the number of loan covenants is negatively related to the creditor rights index. These results suggest that creditor rights significantly influence the contractual terms, perhaps because the level of creditor protection affects the monitoring incentives and/or because it influences the bank's bargaining position.

4.3. Robustness tests: instrumental variable analysis

The potential reverse causality issues may cause the endogeneity problems. In our study, the potential for reverse causality is less of a concern than in pure cross-country analysis because we are examining the impact of creditor rights and the existence of information-sharing agencies on the risk taking of individual bank firms. Nevertheless, we conduct some robustness tests using instrumental variable (IV) analysis. The empirical results are presented in Table 4.

We base the selection of instrumental variables on the theoretical and empirical work in the law, institution, and finance literature (Acemoglu and Johnson, 2005; Beck, Demirgüç-Kunt, and Levine, 2003; Easterly and Levine, 1997; La Porta, Lopez-de-Silanes, Shleifer, and Vishny, 1998, 1999). From the law and finance perspective, La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1999) and Beck, Demirgüç-Kunt, and Levine (2003) show that the historically determined differences in legal traditions help explain international differences in financial systems today. DMS (2007) find a pronounced legal origin effect in credit market institutions. Moreover, legal origin can be thought of as “exogenous” because it was imposed by colonial power in many emerging countries (Acemoglu and Johnson, 2005; La Porta, Lopez-de-Silanes, Shleifer, and Vishny, 1999). Furthermore, the legal origin itself is unlikely to have a direct impact on bank risk taking. Instead, it may exert an indirect impact through the channels of various institutions and regulations. We therefore include legal origin (English, French, German, and Nordic) as instrumental variables for the creditor rights and information sharing measures using data from DMS (2007). According to DMS (2007), the English legal origin includes the common law of England and its former colonies. The French legal origin includes the civil law of France, of countries Napoleon conquered, and of their former colonies. The German legal origin includes the laws of the Germanic countries in central Europe and in East Asia where the German law was transplanted. The Nordic legal origin refers to the laws of the Scandinavian countries. The omitted group is the group of socialist countries. The endowment theory, on the other hand, focuses on the roles of geography and the disease environment in shaping the political and financial institutional development (Acemoglu, Robinson, and Johnson, 2001; Beck, Demirgüç-Kunt, and Levine, 2003). Beck, Demirgüç-Kunt, and Levine BDL (2003) find strong evidence that geographical endowment has substantial impacts on the formation of long-lasting institutions that shape financial development. We therefore follow BDL (2003) in using latitude¹⁸ as an instrumental variable for the creditor rights and information-sharing measures.¹⁹ We also include the ethnic fractionalization²⁰ as

¹⁸ The absolute value of the latitude of the country, scaled to take a value between zero and one, is from La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1999).

¹⁹ We did not use the mortality ratio proposed by Acemoglu and Johnson (2001) because it has a small overlap in countries with the sample we are using.

²⁰ We use the average value of five different indexes of ethnical fractionalization. The data are from Easterly and Levine (1997).

Table 4

Instrumental variables estimation: Z-score and creditor rights.

The dependent variable is log Z-score. The results are based on instrumental variables estimations. Instrumental variables include ethnic fractionalization, latitude, religions, and legal origins. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3	4	5
Creditor rights	−0.470 [0.007]***	−0.484 [0.018]**	−0.488 [0.021]**	−0.492 [0.004]***	−0.448 [0.011]**
Information sharing		1.204 [0.003]***	1.171 [0.024]**		
Information sharing × Creditor rights			0.620 [0.038]**		
Depth of credit information				0.963 [0.009]***	0.974 [0.016]**
Depth of credit information × Creditor rights					0.238 [0.032]**
Diversification index	0.294 [0.008]***	0.295 [0.011]**	0.313 [0.013]**	0.218 [0.057]*	0.279 [0.019]**
Overall capital stringency	0.102 [0.056]*	0.097 [0.061]*	0.127 [0.044]**	0.132 [0.043]**	0.169 [0.031]**
Financial statement transparency	0.252 [0.043]**	0.271 [0.036]**	0.217 [0.069]*	0.249 [0.015]**	0.216 [0.057]*
Deposit insurance coverage	−0.120 [0.006]***	−0.128 [0.004]***	−0.127 [0.009]***	−0.132 [0.007]***	−0.135 [0.005]***
Bank size	0.399 [0.011]**	0.413 [0.006]***	0.410 [0.008]***	0.422 [0.003]***	0.422 [0.006]***
Bank size squared	−0.012 [0.032]**	−0.015 [0.039]**	−0.016 [0.043]**	−0.018 [0.035]**	−0.013 [0.047]**
Too-big-to-fail	−0.331 [0.024]**	−0.363 [0.013]**	−0.357 [0.014]**	−0.343 [0.021]**	−0.315 [0.042]**
HHI	−0.092 [0.180]	−0.144 [0.077]*	−0.145 [0.080]*	−0.116 [0.064]*	−0.117 [0.067]*
Bank growth	−0.082 [0.076]*	−0.075 [0.127]	−0.073 [0.124]	−0.078 [0.128]	−0.084 [0.072]*
Log GDP per capita	0.186 [0.103]	0.259 [0.027]**	0.199 [0.073]*	0.233 [0.047]**	0.239 [0.033]**
Log GDP	0.163 [0.025]**	0.160 [0.033]**	0.117 [0.114]	0.124 [0.073]*	0.138 [0.053]*
Inflation	−0.164 [0.003]***	−0.157 [0.000]***	−0.157 [0.007]***	−0.142 [0.002]***	−0.140 [0.000]***
Constant	1.240 [0.034]**	1.521 [0.030]**	1.610 [0.028]**	1.584 [0.027]**	1.764 [0.024]**
Observations	2363	2363	2363	2363	2363
Adjusted <i>R</i> ²	0.277	0.280	0.280	0.285	0.286
Countries	66	66	66	66	66

an instrumental variable because it has been found that economies with greater ethnic diversity tend to choose institutions that allow those in power to expropriate resources from others (BDL, 2003, 2006). Lastly, it is also reported that a country's cultural heritage, as proxied by religious composition, has a significant impact on shaping its political and financial institutions (La Porta, Lopez-de-Silanes, Shleifer, and Vishny, 1999; Stulz and Williamson, 2003). We therefore include the country's religious composition as an additional instrumental variable.

As can be seen from Table 4, the empirical results are rather robust. The coefficients of creditor rights remain negative and significant, while the coefficients of information-sharing variables remain positive and significant in all model specifications. The results confirm our finding that stronger creditor rights induce more bank risk taking, while information sharing mechanisms help reduce bank risk taking. Consistent with previous results, the coefficients of the interactive term between creditor rights and

information sharing measures are positive and significant across models. Other controls also yield qualitatively similar results. Furthermore, the IV coefficients are somewhat larger than the ordinary least squares (OLS) coefficients, indicating the existence of potential measurement error, which would tend to “attenuate” the coefficient estimate toward zero. It is also worth noting that our main results do not depend on instrumentation, although the latter increases the significance and magnitude of the coefficients.

4.4. Robustness tests: weighted regressions and country level regressions

While the above results suggest that the estimated links between creditor rights, information sharing, and bank risk taking are quite robust, one concern is that the results may be heavily influenced by a subset of large banks in a few key countries. With this issue in mind, the

above results are estimated using heteroskedasticity-robust standard errors. To further alleviate these concerns, we have taken three additional steps.

First, we re-estimate the regressions using weighted OLS regressions (the results are weighted by bank assets). Second, we re-estimate the OLS regressions after eliminating US banks from the sample. These results are summarized in Table 5. While the absolute values of the estimated coefficients are somewhat lower, the main results still strongly hold. Stronger creditor rights are correlated with lower Z-scores, while the measures of information sharing are correlated with higher Z-scores. All of the key coefficients remain statistically different from zero at either the 1% or 5% levels. Moreover, the estimated coefficients and statistical significance of the various control variables remain similar to the results presented in earlier tables.

The third (and more dramatic) approach is to estimate country-level regressions. Here we estimate a single measure of bank risk for each country, and then run a cross-country regression of the 66 countries in our sample. These results are reported in Table 6. In these country-level regressions, we consider three different measures of bank risk: the average log Z-score, the average standard deviation of bank ROA, and the average volatility of banks' net interest margin. In each case, we aggregate the bank risk measures within each country using both an equal-weighted approach and a value-weighted approach (where the banks are weighted by their total assets). Once again, all of the key results are consistently maintained across all risk measures and regardless of weighting. These results provide convincing evidence that our key conclusions regarding creditor rights, information sharing, and risk taking are not

Table 5

Robustness checks: Weighted OLS regressions for all countries and OLS regressions excluding banks in the US.

The dependent variable is log Z-score. $Z\text{-score} = (\text{ROA} + \text{CAR}) / \sigma(\text{ROA})$, where $\text{ROA} = \pi/A$ is return on assets and $\text{CAR} = E/A$ is capital-asset ratio, both averaged over 2000–2007. $\sigma(\text{ROA})$ is the standard deviation of ROA over 2000–2007. Higher Z implies more stability. The estimation is based on weighted OLS regressions for columns 1–5 for all countries. The weights are the bank total assets. For columns 6–10, regressions are OLS for all countries excluding the US. p-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3	4	5	6	7	8	9	10
	Weighted (by bank assets) OLS regressions for all countries					OLS regressions (excluding banks in United States)				
Creditor rights	−0.317 [0.005]***	−0.328 [0.003]***	−0.345 [0.032]**	−0.311 [0.027]**	−0.332 [0.007]***	−0.306 [0.023]**	−0.289 [0.030]**	−0.282 [0.029]**	−0.282 [0.033]**	−0.270 [0.012]**
Information sharing		0.642 [0.007]***	0.634 [0.026]**				0.597 [0.019]**	0.621 [0.002]***		
Information sharing × Creditor rights			0.350 [0.036]**					0.328 [0.037]**		
Depth of credit information				0.629 [0.001]***	0.617 [0.024]**				0.613 [0.008]***	0.605 [0.018]**
Depth of credit information × Creditor rights					0.169 [0.039]**					
Diversification index	0.350 [0.011]**	0.341 [0.041]**	0.256 [0.052]*	0.348 [0.013]**	0.273 [0.057]*	0.473 [0.012]**	0.451 [0.011]**	0.431 [0.013]**	0.376 [0.027]**	0.358 [0.025]**
Overall capital stringency	0.129 [0.048]**	0.137 [0.045]**	0.175 [0.053]*	0.124 [0.069]*	0.151 [0.054]*	0.136 [0.043]**	0.109 [0.055]*	0.129 [0.038]**	0.130 [0.043]**	0.124 [0.036]**
Financial statement transparency	0.225 [0.034]**	0.204 [0.067]*	0.271 [0.029]**	0.224 [0.028]**	0.247 [0.029]**	0.271 [0.036]**	0.287 [0.033]**	0.290 [0.030]**	0.239 [0.054]*	0.248 [0.051]*
Deposit insurance coverage	−0.137 [0.013]**	−0.147 [0.042]**	−0.147 [0.045]**	−0.147 [0.038]**	−0.150 [0.026]**	−0.138 [0.002]***	−0.142 [0.021]**	−0.140 [0.028]**	−0.154 [0.015]**	−0.153 [0.000]***
Bank size	0.219 [0.000]***	0.219 [0.035]**	0.221 [0.037]**	0.206 [0.015]**	0.226 [0.028]**	0.388 [0.000]***	0.401 [0.018]**	0.411 [0.032]**	0.419 [0.012]**	0.426 [0.023]**
Bank size squared	−0.012 [0.017]**	−0.012 [0.015]**	−0.012 [0.014]**	−0.011 [0.024]**	−0.013 [0.018]**	−0.013 [0.015]**	−0.014 [0.014]**	−0.015 [0.033]**	−0.013 [0.035]**	−0.016 [0.028]**
Too-big-to-fail	−0.198 [0.024]**	−0.182 [0.027]**	−0.182 [0.023]**	−0.200 [0.045]**	−0.178 [0.033]**	−0.281 [0.025]**	−0.298 [0.022]**	−0.337 [0.012]**	−0.310 [0.017]**	−0.320 [0.014]**
HHI	−0.110 [0.188]	−0.124 [0.093]*	−0.118 [0.128]	−0.117 [0.163]	−0.122 [0.075]*	−0.083 [0.117]	−0.108 [0.062]*	−0.119 [0.093]*	−0.132 [0.073]*	−0.143 [0.071]*
Bank growth	−0.089 [0.069]*	−0.083 [0.132]	−0.082 [0.133]	−0.088 [0.068]*	−0.081 [0.144]	−0.079 [0.089]*	−0.071 [0.144]	−0.077 [0.091]*	−0.072 [0.198]	−0.073 [0.178]
Log GDP per capita	0.156 [0.058]*	0.165 [0.047]**	0.164 [0.051]*	0.251 [0.026]**	0.170 [0.027]**	0.201 [0.063]*	0.266 [0.046]**	0.258 [0.034]**	0.191 [0.057]*	0.289 [0.011]**
Log GDP	0.152 [0.097]*	0.176 [0.044]**	0.177 [0.041]**	0.169 [0.044]**	0.132 [0.073]*	0.119 [0.034]**	0.127 [0.038]**	0.093 [0.118]	0.118 [0.037]**	0.104 [0.069]*
Inflation	−0.154 [0.002]***	−0.128 [0.004]***	−0.127 [0.003]***	−0.111 [0.009]**	−0.131 [0.004]***	−0.134 [0.000]***	−0.128 [0.000]***	−0.149 [0.000]***	−0.107 [0.000]***	−0.114 [0.000]***
Constant	1.573 [0.019]**	1.695 [0.019]**	1.636 [0.018]**	1.767 [0.034]**	1.405 [0.030]**	1.382 [0.032]**	1.445 [0.031]**	1.547 [0.021]**	1.481 [0.030]**	1.389 [0.034]**
Observations	2386	2363	2363	2363	2363	2029	2006	2006	2006	2006
Adjusted R ²	0.336	0.341	0.341	0.345	0.358	0.217	0.219	0.222	0.230	0.231
Countries	69	66	66	66	66	68	65	65	65	65

Table 6

Country level OLS regressions.

The dependent variables are given in the first row of the table. The estimation is based on OLS regression. *p*-Values are computed by the heteroskedasticity-robust standard errors and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Log Z-score (equal weight)	Log Z-score (asset weight)	σ (ROA) (equal weight)	σ (ROA) (asset weight)	Volatility of net interest margin (equal weight)	Volatility of net interest margin (asset weight)
	1	2	3	4	5	6
Creditor rights	−0.235 [0.015]**	−0.318 [0.006]***	0.0169 [0.003]***	0.0114 [0.027]**	0.0205 [0.024]**	0.0161 [0.041]**
Depth of credit information	0.461 [0.036]**	0.514 [0.013]**	−0.0684 [0.022]**	−0.0406 [0.018]**	−0.0528 [0.019]**	−0.0751 [0.002]***
Depth of credit information × Creditor rights	0.126 [0.025]**	0.147 [0.036]**	−0.0019 [0.011]**	−0.0012 [0.031]**	−0.0011 [0.026]**	−0.0014 [0.016]**
Diversification index	0.404 [0.029]**	0.388 [0.040]**	−0.0043 [0.019]**	−0.0025 [0.016]**	−0.0058 [0.013]**	−0.0020 [0.027]**
Overall capital stringency	0.124 [0.013]**	0.122 [0.018]**	−0.0077 [0.044]**	−0.0058 [0.037]**	−0.0088 [0.061]*	−0.0032 [0.041]**
Financial statement transparency	0.297 [0.058]*	0.243 [0.038]**	−0.0064 [0.033]**	−0.0035 [0.017]**	−0.0082 [0.064]*	−0.0057 [0.035]**
Deposit insurance coverage	−0.149 [0.017]**	−0.102 [0.032]**	0.00076 [0.029]**	0.00047 [0.012]**	0.00084 [0.025]**	0.00066 [0.000]***
HHI	−0.116 [0.108]	−0.102 [0.132]	0.0310 [0.072]*	0.0110 [0.124]	0.0568 [0.086]*	0.0557 [0.218]
Log GDP per capita	0.144 [0.038]**	0.109 [0.069]*	−0.0066 [0.059]*	−0.0038 [0.011]**	−0.0066 [0.049]**	−0.0054 [0.054]*
Log GDP	0.135 [0.059]*	0.164 [0.017]**	−0.0075 [0.024]**	−0.0042 [0.054]*	−0.0063 [0.106]	0.0036 [0.021]**
Inflation	−0.124 [0.018]**	−0.115 [0.012]**	0.00038 [0.000]***	0.00024 [0.000]***	0.00044 [0.016]**	0.00030 [0.000]***
Constant	0.749 [0.031]**	0.684 [0.032]**	0.386 [0.010]**	0.104 [0.002]***	0.834 [0.194]	0.914 [0.009]***
Observations	66	66	66	66	66	66
Adjusted R ²	0.215	0.219	0.280	0.253	0.286	0.271

driven by the weighting of the observations or by the data in a few key countries.

4.5. Using more detailed measures of creditor rights and information sharing

The results reported in Tables 3–6 indicate that there is a strong (and extremely robust) link between bank risk taking and indexes measuring the strength of creditor rights, and the level and depth of information sharing. As a final exercise, we disaggregate these indexes into their subcomponents, to further explore whether specific creditor rights' provisions and/or elements of information sharing have a particularly important influence on bank risk taking.

Table 7 reports regressions where again the dependent variable is the bank's Z-score. We use the same control variables, but now we separately estimate the impact of the four components of the creditor rights index. The results indicate that each of the components has a positive effect on bank risk taking (i.e., a negative effect on the Z-score). Among the four components, the absolute value of the estimated coefficient is largest for the dummy variable indicating whether secured creditors are paid first, and smallest for the dummy variable indicating restrictions on reorganization. The estimated coefficients for the third and fourth components (secured creditors get paid first, and no

management stay) are statistically significant at the 1% level, the coefficient on the second component (no automatic stay in bankruptcy) is significant at the 5% level, while the estimated coefficient on the first component (restrictions on reorganization) is significant at the 10% level. All in all, these results confirm that the various components have a similar effect, and suggest that it is appropriate to use the creditor rights index to proxy for the strength of creditor rights.

Next, we similarly disaggregate the information sharing measure to consider whether it matters if the information sharing takes place through a private bureau or a public registry. These results are summarized in Table 8. Here we find that both sources of information sharing significantly reduce bank risk taking, but the effect is somewhat stronger when the information sharing takes place via a public registry. Once again, the results regarding bank risk taking and creditor rights are robust to both measures of information sharing.

Another potential concern is that the effects of creditor rights depend on not just the codified rights but also the enforcement of those rights. Arguably, a country might have strong creditor rights provisions but if it is too costly in terms of time or money to enforce those rights, then in reality their effect might not be as strong as our index of creditor rights suggests. To address this concern, we ran additional (unreported) regressions where we also controlled for the level of enforcement. We used three

Table 7

Z-score and detailed creditor rights sub-indices.

The dependent variable is log Z-score. The estimation is based on OLS regressions. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3	4
Restrictions on reorganization (cr1)	−0.214 [0.071]*			
No automatic stay (cr2)		−0.272 [0.021]**		
Secured creditor paid first (cr3)			−0.312 [0.008]***	
No management stay (cr4)				−0.246 [0.004]***
Depth of credit information	0.603 [0.006]***	0.594 [0.003]***	0.616 [0.004]***	0.606 [0.008]***
Diversification index	0.342 [0.022]**	0.331 [0.024]**	0.328 [0.040]**	0.312 [0.068]*
Overall capital stringency	0.119 [0.058]*	0.149 [0.043]**	0.145 [0.042]**	0.161 [0.046]**
Financial statement transparency	0.237 [0.028]**	0.237 [0.025]**	0.214 [0.051]*	0.239 [0.037]**
Deposit insurance coverage	−0.150 [0.004]***	−0.142 [0.016]**	−0.153 [0.002]***	−0.150 [0.005]***
Bank size	0.430 [0.003]***	0.441 [0.006]***	0.419 [0.019]**	0.437 [0.006]***
Bank size squared	−0.013 [0.023]**	−0.012 [0.027]**	−0.016 [0.025]**	−0.014 [0.024]**
Too-big-to-fail	−0.283 [0.027]**	−0.288 [0.023]**	−0.268 [0.018]**	−0.285 [0.029]**
HHI	−0.102 [0.066]*	−0.112 [0.090]*	−0.096 [0.064]*	−0.103 [0.060]*
Bank growth	−0.068 [0.071]*	−0.065 [0.138]	−0.064 [0.112]	−0.067 [0.074]*
Log GDP per capita	0.193 [0.016]**	0.153 [0.069]*	0.176 [0.012]**	0.158 [0.057]*
Log GDP	0.093 [0.072]*	0.114 [0.042]**	0.083 [0.075]*	0.104 [0.039]**
Inflation	−0.108 [0.000]***	−0.111 [0.008]***	−0.098 [0.003]***	−0.104 [0.000]***
Constant	1.254 [0.014]**	1.012 [0.018]**	1.608 [0.027]**	1.076 [0.018]**
Observations	2363	2363	2363	2363
Adjusted <i>R</i> ²	0.261	0.262	0.264	0.262
Countries	66	66	66	66

measures to proxy for the time and costs involved in creditor rights' enforcement: (1) The natural log of the average number of procedures involved in enforcing contracts, where a procedure is defined as any direct interaction between the parties, or between the parties and the judge or court officer; (2) the natural log of the average number of calendar days counted from the moment the plaintiff files the lawsuit in court until payment; (3) the average cost of enforcing contracts, (as a percentage of the debt)—these costs include only the official court and enforcement costs and the average attorney fees where the use of an attorney is mandatory or common. These proxies were first developed by Djankov, La Porta, Lopez-de-Silanes, and Shleifer (2003), and have been updated in the World Bank's "Doing Business" database. As an additional proxy for creditor rights enforcement, we also controlled for the efficiency of the judicial system, as measured by La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998).

In these unreported regressions, we ran four additional regressions that controlled separately for each of the enforcement proxies, and a final regression where we simultaneously controlled for each of the proxies. In each case, none of the enforcement proxies were significant (at the 10% level), and in each case, the main results regarding creditor rights, information sharing, and bank risk taking were upheld.²¹

²¹ While the enforcement variables did not have a statistically significant effect on bank risk taking, it is interesting to note that the efficiency of the judicial system is positively associated with the Z-score (i.e., negatively associated with bank risk taking). Each of the other measures (log number of procedures, log time to enforce contracts, and the cost of enforcing contracts) was negatively associated with the Z-score (i.e., positively associated with bank risk taking). Overall, these findings suggest that efficient enforcement tends to reduce the bank risk taking but the effects are not statistically significant. The results are therefore not reported for brevity but available from the authors.

Table 8

Robustness Tests: Z-score and more information sharing measures.

The dependent variable is log Z-score. The estimation is based on OLS regressions. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3	4
	No interaction term		With interaction terms	
Creditor rights	−0.207 [0.036]**	−0.323 [0.037]**	−0.285 [0.022]**	−0.317 [0.010]**
Private bureau	0.458 [0.012]**		0.476 [0.024]**	
Public registry	0.651 [0.003]***		0.679 [0.009]***	
Private bureau coverage		0.176 [0.029]**		0.168 [0.034]**
Public registry coverage		0.285 [0.015]**		0.273 [0.029]**
Private bureau × Creditor rights			0.227 [0.037]**	
Public registry × Creditor rights			0.342 [0.018]**	
Private bureau coverage × Creditor rights				0.103 [0.039]**
Public registry coverage × Creditor rights				0.154 [0.021]**
Diversification index	0.370 [0.014]**	0.341 [0.027]**	0.369 [0.015]**	0.305 [0.053]*
Overall capital stringency	0.093 [0.051]*	0.136 [0.034]**	0.124 [0.049]**	0.148 [0.026]**
Financial statement transparency	0.229 [0.043]**	0.294 [0.036]**	0.206 [0.052]*	0.309 [0.019]**
Deposit insurance coverage	−0.147 [0.008]***	−0.137 [0.012]**	−0.150 [0.023]**	−0.140 [0.000]***
Bank size	0.426 [0.000]***	0.422 [0.000]***	0.415 [0.000]***	0.391 [0.000]***
Bank size squared	−0.014 [0.015]**	−0.014 [0.008]***	−0.013 [0.021]**	−0.013 [0.011]**
Too-big-to-fail	−0.288 [0.019]**	−0.266 [0.033]**	−0.308 [0.014]**	−0.285 [0.022]**
HHI	−0.148 [0.095]*	−0.110 [0.076]*	−0.170 [0.087]*	−0.137 [0.065]*
Bank growth	−0.061 [0.127]	−0.069 [0.094]*	−0.068 [0.096]*	−0.063 [0.125]
Log GDP per capita	0.243 [0.002]**	0.192 [0.059]*	0.236 [0.013]**	0.214 [0.036]**
Log GDP	0.103 [0.054]*	0.096 [0.126]	0.099 [0.063]*	0.122 [0.042]**
Inflation	−0.119 [0.000]***	−0.112 [0.000]***	−0.124 [0.006]***	−0.101 [0.003]***
Constant	1.008 [0.021]**	1.902 [0.011]**	1.804 [0.024]**	2.016 [0.018]**
Observations	2363	2363	2363	2363
Adjusted R^2	0.257	0.260	0.260	0.273
Countries	66	66	66	66

5. Do creditor rights protection and information sharing affect the likelihood of financial crisis and economic growth?

The results reported in Section 4 clearly indicate that the strength of creditor rights protection in bankruptcy and the level of information sharing both have a profound effect on the level of bank risk taking. While these results are interesting, they leave open a very important question, which is whether higher bank risk taking is good or bad. While it is well beyond the scope of this paper to

address the question of what is the optimal level of bank risk taking across different countries, we can hopefully shed some insight on the consequences of higher (and lower) risk taking. With this point in mind, we next consider two possible consequences of greater risk taking. First, from a negative perspective, higher bank risk taking may increase the likelihood of a financial crisis emerging within a given country. From a more positive perspective, greater risk taking may translate into banks providing more credit to a wider range of borrowers, which in turn leads to higher economic growth.

5.1. Creditor rights, information sharing, and banking crisis

To test whether the strength of creditor rights and information sharing affect banking system fragility, we follow Demirgüç-Kunt and Detragiache (2002) and Beck, Demirgüç-Kunt, and Levine (2006) and estimate the following logit probability model:

$$Prob(crisis_{jt} = 1|X_{jt}) = \frac{\exp(X_{jt}\beta)}{1 + \exp(X_{jt}\beta)}, \quad (2)$$

where $crisis_{jt}$ is a dummy variable that takes the value of one when a systemic banking crisis occurs in country j and time t , and a value of zero otherwise. X_{jt} is a vector of the key variables such as the creditor rights index, measures of information sharing, and the interaction of these key measures. X_{jt} also includes other control variables such as bank concentration, deposit insurance coverage, and macro variables such as real GDP growth, terms of trade change, real interest rate, inflation, M2 to foreign reserves ratio, depreciation of domestic currency, credit growth, log GDP per capita, and log GDP. β is a vector of parameters. All the independent variables (including the creditor rights and information sharing measures) are time varying and are lagged by two time periods to avoid simultaneity issues. The time-varying measures of creditor rights and information sharing are from DMS (2007).

To explore this issue, we take advantage of all available data from the Honohan and Laeven (2005) and Laeven and Valencia (2008) studies, and extend our sample to banks in 93 countries over a 28-year sample period between 1980 and 2007. The results are reported in Table 9. Relative to the results reported in Tables 2–8, which focused on individual bank risk taking, these results are estimated on a country-wide level. Since this eliminates the need to control for individual bank measures, we are able to capture data from a wider number of countries over a longer period of time. The sample period ends in 2007 corresponding with the endpoint of Laeven and Valencia's crisis series.

Across all specifications, stronger creditor rights are correlated with a greater likelihood of financial crisis. The results presented in Section 4 suggest a primary channel where stronger creditor rights induce greater bank risk taking, which in turn raises the likelihood of crisis. Specifically, a one standard deviation increase in *Creditor rights* (1.203) increases the likelihood of a systemic banking crisis by about 4.8%.

The results reported in Table 9 also indicate that the greater information sharing significantly reduces the likelihood of a financial crisis. Specifically, the presence of an information sharing agency will reduce the likelihood of a systemic banking crisis by about 4.8%. This finding is consistent with earlier results that suggest that information sharing significantly reduces bank risk taking, and it further confirms what appears to be the universal benefits of information sharing. Looking more closely at the results in Table 9, we see that information sharing significantly weakens the positive link between creditor rights and banking crisis.

As discussed earlier, we also test the robustness of the results using the four theory-based indicators of systemic

bank shocks (SBS) proposed by Boyd, De Nicolo, and Loukoianova (2009). The empirical results are reported in columns 3–6 in Table 9. As can be seen from the table, the empirical results are highly consistent with our previous findings. We find robust evidence that creditor rights increase the likelihood of a banking crisis, while information sharing reduces the likelihood of a banking crisis and weakens the positive link between creditor rights and the likelihood of a banking crisis.

5.2. Creditor rights, information sharing, and industry growth

From an outcome-based perspective, we follow the pioneering work by Rajan and Zingales (1998) to test the effects of creditor rights and information sharing on industry growth as well as the moderating effect of external finance dependence. The methodology is widely used in the recent literature (e.g., Cetorelli and Gambera, 2001; Beck and Levine, 2002; Kroszner, Laeven, and Klingebiel, 2007). Specifically, we estimate the following model:

$$Growth_{ik} = f(\text{share}_{ik}, Y_{ik}, EFD_i \times Y_{ik}, \text{country indicators, industry indicators}) + e_{ik}, \quad (3)$$

where $Growth_{ik}$ is the growth rate in real value added for 1996–2005 for industry i in country k (i.e., change in the log of real value added in that industry between 1996 and 2005),²² $share_{ik}$ is the industry i 's share of total value added in manufacturing in 1996, and Y_{ik} is a vector of key variables such as the creditor rights index, measure of information sharing, and their interaction term. Y_{ik} also includes other control variables such as bank concentration, deposit insurance coverage, and financial market development indicators such as private credit to GDP ratio and stock market capitalization to GDP ratio. EFD_i measures the external finance dependence of industry i defined as the fraction of capital expenditures not financed with internal funds (using US firms in the same industry between 1980 and 1990 as the benchmark, (Rajan and Zingales, 1998), and e_{ik} is the error term. The sample excludes the industrial sectors in the US, which serves as the benchmark (Rajan and Zingales, 1998). The empirical results are presented in columns 1–5 in Table 10. The IV regression results using a similar set of instruments are presented in columns 6–10 in Table 10.

Looking at Table 10, we find that stronger creditor rights is associated with faster industry growth, as indicated by the positive and statistically significant coefficients on the *Creditor rights index* across various model specifications. The results suggest that although creditor rights induce more bank risk taking, they enhance industry growth by encouraging more credit availability to the private sector. Consistent with our previous findings, we also see that information sharing is correlated with higher sector growth. This result holds for each of the information sharing proxies (*Information sharing* and *Depth of credit information*). More

²² Following Rajan and Zingales (1998), we dropped observations that have revenue growth measure less than -1 and above $+1$ to reduce the impact of outliers. Less than 1% of the observations are affected.

Table 9

Information sharing, creditor rights and banking crisis.

The logit probability model is estimated. The sample period is 1980–2007. The dependent variable is a crisis dummy that takes on the value of one if there is a systemic banking crisis, and the value of zero otherwise. The banking crisis dummy in columns 1 and 2 is defined by Honohan and Laeven (2005) and Laeven and Valencia (2008). The banking crisis dummies in columns 3–6 are defined by Boyd, De Nicolò, and Loukoianova (2009). SBSL10 and SBSL25 are two bank crisis indicators proposed by Boyd, De Nicolò, and Loukoianova (2009) that are related to sharp decreases in total loans. They equal one if real domestic lending growth is lower than the 10th and 25th percentile of the entire distribution of real domestic bank credit growth across countries, respectively. SBSL10 and SBSL25 are two bank crisis indicators also proposed by Boyd, De Nicolò, and Loukoianova (2009) that measure a sharp decrease in total bank deposits. They equal one if the growth rate of the deposit-to-GDP ratio is lower than the 10th and 25th percentile of its distribution across countries, respectively. All the independent variables are lagged two periods. Growth is the rate of growth of real GDP. Terms of trade equals the changes in the terms of trade of goods and services. Real interest rate is the nominal interest rate minus the contemporaneous rate of inflation. Inflation is the rate of change of the GDP deflator. M2/reserves is the ratio of M2 to foreign exchange reserves of the central bank. Credit growth is the real growth of domestic credit. Depreciation of domestic currency is the rate of change of the exchange rate. Deposit insurance coverage is the ratio of deposit insurance coverage to deposits per capita. Bank concentration is calculated as the fraction of assets held by the three largest banks in each country, averaged over the sample period. Log GDP per capita is logarithm of real GDP per capita in USD. Log GDP is log real GDP in USD. The marginal effects (dy/dx) of the logit regressions are presented. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3 SBSL10	4 SBSL25	5 SBSL10	6 SBSL25
Creditor rights	0.042 [0.022]**	0.040 [0.025]**	0.043 [0.014]**	0.050 [0.019]**	0.044 [0.018]**	0.051 [0.013]**
Information sharing	−0.047 [0.029]**	−0.048 [0.012]**	−0.049 [0.014]**	−0.053 [0.017]**	−0.046 [0.025]**	−0.051 [0.011]**
Information sharing × Creditor rights		−0.056 [0.028]**	−0.034 [0.026]**	−0.036 [0.035]**	−0.040 [0.021]**	−0.057 [0.024]**
Bank concentration	0.060 [0.154]	0.053 [0.061]*	0.071 [0.082]*	0.084 [0.074]*	0.069 [0.067]*	0.080 [0.121]
Deposit insurance coverage	0.018 [0.000]***	0.018 [0.000]***	0.017 [0.012]**	0.022 [0.026]**	0.013 [0.016]**	0.018 [0.015]**
Real GDP growth	−0.0040 [0.003]***	−0.0047 [0.002]***	−0.0032 [0.014]**	−0.0041 [0.007]***	−0.0030 [0.006]***	−0.0045 [0.008]***
Terms of trade change	−0.00054 [0.032]**	−0.00052 [0.115]	−0.0005 [0.076]*	−0.0006 [0.122]	−0.0004 [0.146]	−0.0008 [0.048]**
Real interest rate	0.0002 [0.037]**	0.0002 [0.023]**	0.0001 [0.030]**	0.0002 [0.032]**	0.0001 [0.031]**	0.0003 [0.022]**
Inflation	0.0000 [0.121]	0.0000 [0.046]**	0.0001 [0.027]**	0.0001 [0.104]	0.0001 [0.101]	0.0001 [0.089]*
M2/reserves	0.0050 [0.025]**	0.0052 [0.071]*	0.0033 [0.195]	0.0051 [0.046]**	0.0039 [0.109]	0.0055 [0.097]*
Depreciation of domestic currency	0.00047 [0.051]*	0.00044 [0.044]**	0.00027 [0.038]**	0.00034 [0.172]	0.00021 [0.127]	0.00032 [0.071]*
Credit growth	0.0024 [0.001]***	0.0020 [0.002]***	0.003 [0.024]**	0.005 [0.021]**	0.002 [0.079]*	0.005 [0.015]**
Log GDP per capita	−0.046 [0.002]***	−0.055 [0.003]***	−0.048 [0.036]**	−0.032 [0.044]**	−0.039 [0.033]**	−0.068 [0.035]**
Log GDP	−0.024 [0.118]	−0.024 [0.102]	−0.031 [0.147]	−0.054 [0.068]*	−0.021 [0.102]	−0.033 [0.174]
Observations	2516	2516	2,516	2,516	2,516	2,516
No. of crises	380	380	251	629	251	629
Pseudo R ²	0.166	0.164	0.115	0.171	0.136	0.154
Log_likelihood	−979.7	−977.9	−670	−1,198	−702	−1,259
Countries	93	93	93	93	93	93

importantly, we find that creditor rights and information sharing help externally dependent industries grow faster, as indicated by the positive and statistically significant interaction terms between *Creditor rights* and *EFD* and between *Information sharing* and *EFD*. Moreover, we find that information sharing helps strengthen the positive link between creditor rights and industrial growth, and the effect is stronger in the financially dependent industries. Regarding the control variables, we find that banking concentration and deposit insurance coverage are negatively associated with industrial growth. Alternatively, higher levels of stock market capitalization, and bank credit to private sectors are positively associated with industrial growth. All of these effects are stronger in the financially

dependent industries. The results echo the findings in previous studies (e.g., Rajan and Zingales, 1998; Beck and Levine, 2002).²³

²³ Following the literature (e.g., Beck, Levine, and Loayza, 2000; Beck and Levine, 2004), we also explore the impacts of creditor rights and information sharing on real per capita GDP growth across the sample countries over the past 30 years using dynamic panel regressions. Specifically, the regressions are based on two-step system generalized method of moments (GMM) panel estimations (Arellano and Bover, 1995), with data averaged over six five-year periods from 1978–2007, and using lagged values as instruments. We find the empirical results highly robust. Stronger creditor rights and better information sharing are associated with a higher economic growth rate. For brevity, the empirical results are not reported but available from the authors.

Table 10

Creditor rights, information sharing, and industry growth.

The dependent variable is the growth rate in real value added for 1996–2005 for each industry in each country, using the data from UNIDO INDSTAT4, 2008. The sample excludes the industrial sectors in the US, which serves as the benchmark (Rajan and Zingales, 1998). In regressions 1–5, estimation is by ordinary least squares. In regressions 6–10, instrument variable estimations are applied. Instrumental variables include ethnic fractionalization, latitude, religions, and legal origins. Country and industry dummy variables are included in the regressions but not reported. Share in value added is the industry's share of total value added in manufacturing in 1996. EFD is external finance dependence [the fraction of capital expenditures not financed with internal funds for US firms in the same industry between 1980 and 1990 (Rajan and Zingales, 1998)]. Private credit is private credit divided by GDP in 1996. Market cap is stock market capitalization divided by GDP in 1996. Total capitalization is the sum of the private credit and stock market capitalization variables. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. ***, **, * Denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	OLS					IV				
	1	2	3	4	5	6	7	8	9	10
Share in value added	−0.2078 [0.018]**	−0.1990 [0.021]**	−0.2115 [0.026]**	−0.2228 [0.027]**	−0.2365 [0.023]**	−0.2845 [0.003]***	−0.2559 [0.006]***	−0.2632 [0.021]**	−0.2324 [0.027]**	−0.2286 [0.019]**
Creditor rights	0.0114 [0.001]***	0.0106 [0.030]**	0.0158 [0.000]***	0.0114 [0.025]**	0.0103 [0.000]***	0.0252 [0.043]**	0.0231 [0.045]**	0.0264 [0.012]**	0.0229 [0.011]**	0.0225 [0.042]**
EFD × Creditor rights	0.0128 [0.040]**	0.0101 [0.048]**	0.0169 [0.046]**	0.0123 [0.044]**	0.0138 [0.008]***	0.0249 [0.029]**	0.0273 [0.004]***	0.022 [0.026]**	0.029 [0.021]**	0.0219 [0.039]**
Information sharing		0.0270 [0.000]***	0.0284 [0.000]***				0.0413 [0.017]**	0.0402 [0.032]**		
EFD × Information sharing		0.0201 [0.011]**	0.0223 [0.045]**				0.0305 [0.048]**	0.0361 [0.028]**		
Creditor rights × Information sharing			0.0115					0.0198		
EFD × Creditor rights × Information sharing			[0.013]** 0.0155					[0.007]*** 0.0187		
Depth of information sharing			[0.043]**	0.0286 [0.024]**	0.0220 [0.015]**			[0.029]**	0.0394 [0.013]**	0.0389 [0.007]***
EFD × Depth of information sharing				0.0014	0.0016				0.0025	0.0027
Creditor rights × Depth of information sharing				[0.008]***	[0.043]** 0.0073				[0.016]**	[0.032]** 0.0091
EFD × Creditor rights × Depth of information sharing					[0.039]** 0.0062					[0.003]*** 0.0076
HHI	−0.1488 [0.012]**	−0.1590 [0.014]**	−0.1413 [0.051]*	−0.1336 [0.017]**	−0.1009 [0.077]*	−0.2478 [0.013]**	−0.2897 [0.012]**	−0.2663 [0.013]**	−0.2976 [0.057]*	−0.3473 [0.026]**
EFD × HHI	−0.0498 [0.034]**	−0.0544 [0.031]**	−0.0525 [0.059]*	−0.0607 [0.023]**	−0.0680 [0.018]**	−0.0422 [0.047]**	−0.0434 [0.052]*	−0.0549 [0.064]*	−0.0643 [0.022]**	−0.0392 [0.097]*
Deposit insurance coverage	−0.00033 [0.057]*	−0.00051 [0.023]**	−0.00045 [0.011]**	−0.00050 [0.014]**	−0.00043 [0.016]**	−0.00040 [0.021]**	−0.00038 [0.047]**	−0.00035 [0.070]*	−0.00046 [0.018]**	−0.00053 [0.057]*
EFD × Deposit insurance coverage	−0.00044	−0.00043	−0.00049	−0.00048	−0.00036	−0.00021	−0.00022	−0.00032	−0.00038	−0.00027
Market cap	[0.039]** 0.1642	[0.036]** 0.0981	[0.058]* 0.1031	[0.031]** 0.1554	[0.042]** 0.1611	[0.066]* 0.1649	[0.026]** 0.1504	[0.017]** 0.1453	[0.014]** 0.1540	[0.035]** 0.1549
EFD × Market cap	[0.064]* 0.0297	[0.054]* 0.0255	[0.024]** 0.0288	[0.063]* 0.0301	[0.037]** 0.0336	[0.031]** 0.0269	[0.046]** 0.0275	[0.057]* 0.0296	[0.044]** 0.0228	[0.132] 0.0307
Private credit	[0.019]** 0.0159	[0.078]* 0.0135	[0.019]** 0.0122	[0.058]* 0.0186	[0.014]** 0.0133	[0.062]* 0.0225	[0.014]** 0.0259	[0.053]* 0.0238	[0.039]** 0.0232	[0.016]** 0.0207
EFD × Private credit	[0.003]*** 0.0195	[0.007]*** 0.0122	[0.040]** 0.0121	[0.054]* 0.0111	[0.036]** 0.0140	[0.026]** 0.0275	[0.030]** 0.0249	[0.027]** 0.0271	[0.061]* 0.0272	[0.049]** 0.0214
Constant	[0.016]** −1.8209	[0.027]** −1.7701	[0.028]** −1.4661	[0.049]** −0.8734	[0.031]** −0.7071	[0.006]*** −1.3152	[0.013]** −1.2812	[0.025]** −1.4527	[0.008]*** −0.5989	[0.041]** −0.6754
	[0.000]***	[0.000]***	[0.003]***	[0.043]**	[0.000]***	[0.020]**	[0.011]**	[0.056]*	[0.023]**	[0.070]*
Observations	2430	2430	2430	2430	2430	2430	2430	2430	2430	2430
Countries	79	79	79	79	79	79	79	79	79	79
Adjusted R ²	0.366	0.375	0.323	0.369	0.366	0.274	0.299	0.266	0.280	0.290

6. Conclusion

A large literature looking at cross-country economic performance highlights the importance of bank lending

(see, for example, Levine, 2005). Given this importance, there is widespread interest in understanding the factors that encourage or discourage banks from providing credit. At the same time, the recent financial crisis has generated

renewed interest into how the institutional environment and regulatory environment influence bank risk taking.

Within this backdrop, our paper explores how creditor rights and information sharing influence the level of bank risk taking, the likelihood of financial crisis, and economic growth. While we believe our paper is the first to address this specific issue, ours is part of a growing literature emphasizing the important role of creditor rights and information sharing (e.g., Pagano and Jappelli, 1993; Padilla and Pagano, 1997; Jappelli and Pagano, 2002; Djankov, McLiesh, and Shleifer, 2007; John, Litov, and Yeung, 2008; Acharya, Amihud, and Litov, 2009).

Looking at a sample of nearly 2,400 banks in 69 countries, we find that stronger creditor rights tend to promote greater risk-taking. Consistent with this finding, we also find that stronger creditor rights increase the likelihood of that country experiencing a financial crisis. On a more positive note, we find that stronger creditor rights are correlated with higher growth. In this regard, we find that there is both a “bright” side and a “dark” side to stronger creditor rights.

We hasten to point out that our notion of a dark side is very different from that described recently by Acharya, Amihud, and Litov (2009). Their study instead focuses on corporate risk taking, and they show that stronger creditor rights induce managers to engage in risk-reducing strategies such as diversifying acquisitions. In this regard, their concern is that stronger creditor rights may cause individual corporate managers to take on insufficient risk, which in turn may lead to reduced

economic growth. Rather than focusing on the behavior of corporate borrowers, our study focuses on lenders and examines the willingness of banks (the chief provider of capital in many markets) to take on risk. Putting our results together suggests that stronger creditor rights have a differential effect on the risk-taking incentives of borrowers and lenders.

In contrast to the mixed results we find regarding the effects of stronger creditor rights, we find that the benefits of information sharing appear to be universally positive. Greater information sharing leads to higher bank profitability, lower bank risk, a reduced likelihood of financial crisis, and higher economic growth. These findings lend further support to the argument that there are global benefits to increased transparency and information sharing.

Appendix A

See Table A1.

Appendix B

See Table B1.

Appendix C

See Table C1.

Table A1
Variable definitions and data sources.

Variable	Definition	Original sources
<i>Bank-level data</i>		
Log Z-score	Equals $\log \text{ of } (\text{ROA} + \text{CAR}) / \sigma(\text{ROA})$, where $\text{ROA} = \pi/A$ is return on assets and $\text{CAR} = E/A$ is capital-asset ratio, both averaged over 2000–2007. $\sigma(\text{ROA})$ is the standard deviation of ROA over 2000–2007. Higher Z implies more stability.	BankScope
ROA	Return on assets, averaged over 2000–2007. Higher value implies more stability.	BankScope
CAR	Capital-asset ratio, averaged over 2000–2007. Higher value implies more stability.	BankScope
$\sigma(\text{ROA})$	Equals the standard deviation of ROA of a bank, computed over the period 2000–2007.	BankScope
Volatility of net interest margin	Standard deviation of net interest margin over the period 2000–2007, where $\text{net interest margin} = 100 \times \text{net interest income} / \text{earning assets}$.	BankScope
Bank growth	Total revenue growth rate of a bank, averaged over 2000–2007.	BankScope
Too-big-to-fail	A dummy variable that takes a value of one if the bank's share in the country's total deposits exceeds 10%.	BankScope
Bank size	Natural logarithm of total assets, averaged over 2000–2007.	BankScope
Non-performing loans (%)	Non-performing loans divided by total loans, in percentage, averaged over 2000–2007.	BankScope
Loan loss provisions (%)	Loan loss provisions divided by total loans, in percentage, averaged over 2000–2007.	BankScope
<i>Country-level data</i>		
HHI	To control for competition, we use the Herfindahl–Hirschman Index, defined as the sum of the squared shares of bank deposits to total deposits within a given country, averaged over the period 2000 to 2007.	BankScope
Creditor rights (cr)	An index aggregating creditor rights, following La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998). It is the sum of the four indexes that follow (cr1, cr2, cr3, and cr4). It ranges between 0 and 4 over the period of 1978–2003.	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998), Djankov, McLiesh, and Shleifer (2007)

Table A1 (continued)

Variable	Definition	Original sources
Restrictions on reorganization (cr1)	Equals one if the reorganization procedure imposes restrictions, such as creditors' consent or minimum dividend for a debtor to be able to file for reorganization. It equals zero for countries without such restrictions.	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998), Djankov, McLiesh, and Shleifer (2007)
No automatic stay (cr2)	Equals one if the reorganization procedure does not impose an automatic stay on the assets of the firm upon filing the reorganization petition and creditors are able to seize their collateral after the reorganization petition is approved. It equals zero if such a restriction does exist in the law.	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998), Djankov, McLiesh, and Shleifer (2007)
Secured creditor paid first (cr3)	Equals one if secured creditors are ranked first in the distribution of the proceeds that result from the disposition of the assets of a bankrupt firm, as opposed to other creditors such as employees or government. Equals zero if non-secured creditors, such as the government and workers, are given absolute priority.	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998), Djankov, McLiesh, and Shleifer (2007)
No management stay (cr4)	Equals one if an official appointed by the court, or by the creditors, is responsible for the operation of the business during reorganization, that is, management does not retain administration of its property pending the resolution of the reorganization. Equivalently, this variable equals one if the debtor does not keep the administration of its property pending the resolution of the reorganization process, and zero otherwise.	La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998), Djankov, McLiesh, and Shleifer (2007)
Depth of credit information	An index that measures the information contents of the credit information. A value of one is added to the index when a country's information agencies have each of these characteristics: (1) both positive credit information (for example, loan amounts and pattern of on-time repayments) and negative information (for example, late payments, number and amount of defaults, and bankruptcies) are distributed; (2) data on both firms and individual borrowers are distributed; (3) data from retailers, trade creditors, or utilities, as well as from financial institutions, are distributed; (4) more than two years of historical data are distributed; (5) data are collected on all loans of value above 1% of income per capita; and (6) laws provide for borrowers' right to inspect their own data. The index ranges from 0–6, with higher values indicating the availability of more credit information, from either a public registry or a private bureau, to facilitate lending decisions.	Djankov, McLiesh, and Shleifer (2007), World Bank "Doing Business" database
Information sharing	A dummy variable that equals one if an information sharing agency (public registry or private bureau) operates in the country during the sample period, zero otherwise.	Djankov, McLiesh, and Shleifer (2007), World Bank "Doing Business" database
Public registry	A dummy variable that equals one if a public registry operates in the country during the sample period, zero otherwise.	Djankov, McLiesh, and Shleifer (2007), World Bank "Doing Business" database
Private bureau	A dummy variable that equals one if a private bureau operates in the country during the sample period, zero otherwise.	Djankov, McLiesh, and Shleifer (2007), World Bank "Doing Business" database
Public registry coverage	The number of individuals and firms listed in a public credit registry with information on repayment history, unpaid debts, or credit outstanding from the past five years scaled by adult population, averaged over 2004–2007.	Djankov, McLiesh, and Shleifer (2007), World Bank "Doing Business" database
Private bureau coverage	The number of individuals and firms listed by a private credit bureau with information on repayment history, unpaid debts, or credit outstanding from the past five years scaled by the adult population, averaged over 2004–2007.	Djankov, McLiesh, and Shleifer (2007), World Bank "Doing Business" database
Diversification index	Whether there are explicit, verifiable, quantifiable guidelines for asset diversification and banks are allowed to make loans abroad. Higher values indicate more diversification.	Barth, Caprio, and Levine (2006)
Overall capital stringency	Whether the capital requirement reflects certain risk elements and deducts certain market value losses from capital adequacy is determined. Higher values indicate greater stringency.	Barth, Caprio, and Levine (2006)
Financial statement transparency	The transparency of bank financial statement practices. Higher values indicate better transparency.	Barth, Caprio, and Levine (2006)
Deposit insurance coverage	Log of one plus ratio of deposit insurance coverage to deposits per capita	Demirgüç-Kunt, Kane, and Laeven (2008)
Inflation	Percentage inflation rate, GDP deflator.	World Development Indicators (WDI)
Log GDP per capita	Log real GDP per capita, in US dollars.	World Development Indicators (WDI)
Log GDP	Natural logarithm of the real GDP, in US dollars.	World Development Indicators (WDI)
GDP volatility	Standard deviation of real GDP growth rates over 2000–2007.	
Voice and accountability	The indicator measures the extent to which a country's citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association, and free media. The value of year 2005 is used in this study. Higher values mean greater political rights.	Kaufmann, Kraay, and Mastruzzi (2006)

Table A1 (continued)

Variable	Definition	Original sources
Government effectiveness	The indicator measures the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies. The value of year 2005 is used in this study. Higher values mean higher quality of public and civil service.	Kaufmann, Kraay, and Mastruzzi (2006)
Rule of law	The indicator measures the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, the police, and the courts, as well as the likelihood of crime and violence. The value of year 2005 is used in this study. Higher values indicate stronger law and order.	Kaufmann, Kraay, and Mastruzzi (2006)
Political stability	The indicator measures the perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means, including political violence and terrorism. The value of year 2005 is used in this study. Higher values mean more stable political environment.	Kaufmann, Kraay, and Mastruzzi (2006)
Quality of regulation	The indicator measures the ability of the government to formulate and implement sound policies and regulations that permit and promote market competition and private-sector development. The value of year 2005 is used in this study. Higher values mean higher quality of regulation.	Kaufmann, Kraay, and Mastruzzi (2006)
Control of corruption	The indicator measures the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as "capture" of the state by elites and private interests. The value of year 2005 is used in this study. Higher values indicate better control of corruption.	Kaufmann, Kraay, and Mastruzzi (2006)
Transparency index	It is based on over 20 independent sources and comprises an aggregate transparency index with two subcomponents: economic/institutional transparency, and political transparency. A high value of the index indicates easy public access to relevant and reliable information of markets, and legal and political systems.	Kaufmann and Bellver (2005)
Systemic banking crisis dummy	It takes on the value of one if there is a systemic banking crisis, and the value of zero otherwise.	Honohan and Laeven (2005)
Terms of trade	The changes in the terms of trade of goods and services.	Beck, Demirgüç-Kunt, and Levine (2006), World Development Indicators (WDI)
Real interest rate	The nominal interest rate minus the contemporaneous rate of inflation.	Beck, Demirgüç-Kunt, and Levine (2006), World Development Indicators (WDI)
M2/reserves	The ratio of M2 to foreign exchange reserves of the central bank.	Beck, Demirgüç-Kunt, and Levine (2006), World Development Indicators (WDI)
Credit growth	The real growth of domestic credit.	Beck, Demirgüç-Kunt, and Levine (2006), World Development Indicators (WDI)
Depreciation of domestic currency	The rate of change of the exchange rate.	Beck, Demirgüç-Kunt, and Levine (2006), World Development Indicators (WDI)
Bank concentration	The fraction of assets held by the three largest banks in each country, averaged over the sample period.	Beck, Demirgüç-Kunt, and Levine (2006), BankScope
Private credit	Private credit divided by GDP in 1996.	Beck, Levine, and Loayza (2000), updated in 2007
Market capitalization	Stock market capitalization divided by GDP in 1996.	Beck, Levine, and Loayza (2000), updated in 2007
<i>Industrial-level data</i>		
External finance dependence	The fraction of capital expenditures not financed with internal funds for US firms in the same industry between 1980 and 1990.	Rajan and Zingales (1998)
Growth rate in real value added	The growth rate in real value added for 1996–2005 for each industry in each country. The sample excludes the industrial sectors in the US, which serves as the benchmark (Rajan and Zingales, 1998).	UNIDO INDSTAT4, 2008
Share in value added	The industry's share of total value added in manufacturing in 1996 for each industry in each country.	UNIDO INDSTAT4, 2008

Table B1Panel estimations: Log Z-score, ROA, CAR, and $\sigma(\text{ROA})$.

The dependent variable is the logarithm of Z-score, which is constructed based on three-year periods over 1996–2007. The sample period has a total of 12 years and provides four three-year non-overlapping subperiods. The estimation is based on panel country fixed-effects estimations. In all estimations, time and country fixed effects are included but not reported. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	1	2	3
Creditor rights	−0.273 [0.019]**	−0.264 [0.020]**	−0.252 [0.018]**
Information sharing		0.774 [0.013]**	0.640 [0.026]**
Information sharing × Creditor rights			0.547 [0.041]**
Diversification index	0.125 [0.014]**	0.128 [0.013]**	0.127 [0.011]**
Overall capital stringency	0.147 [0.015]**	0.186 [0.012]**	0.190 [0.014]**
Financial statement transparency	0.216 [0.032]**	0.215 [0.056]*	0.219 [0.038]**
Deposit insurance coverage	−0.187 [0.034]**	−0.189 [0.025]**	−0.157 [0.036]**
Bank size	0.651 [0.008]***	0.539 [0.008]***	0.535 [0.007]***
Bank size squared	−0.028 [0.015]**	−0.022 [0.010]***	−0.023 [0.011]**
Too-big-to-fail	−0.332 [0.023]**	−0.341 [0.022]**	−0.336 [0.021]**
HHI	−0.051 [0.070]*	−0.055 [0.068]*	−0.053 [0.057]*
Bank growth	−0.074 [0.069]*	−0.073 [0.059]*	−0.069 [0.066]*
Log GDP per capita	0.388 [0.034]**	0.369 [0.053]*	0.342 [0.012]**
Log GDP	0.124 [0.037]**	0.122 [0.039]**	0.144 [0.063]*
Inflation	−0.114 [0.058]*	−0.107 [0.032]**	−0.111 [0.033]**
Observations	6,793	6,793	6,315
Countries	66	66	66
Banks	4,046	4,046	4,046
Adjusted R ²	0.261	0.263	0.264

Table C1

Robustness tests: Z-score, creditor rights, transparency, and additional controls.

The dependent variable is log Z-score. The estimation is based on OLS regressions. *p*-Values are computed by the heteroskedasticity-robust standard errors clustered for countries and are presented in brackets. *, **, *** Represent statistical significance at the 10%, 5%, and 1% levels, respectively. Transparency index is from Kaufmann and Bellver (2005). It is based on over 20 independent sources and comprises an aggregate transparency index with two subcomponents: economic/institutional transparency, and political transparency. A high value of the index indicates easy public access to relevant and reliable information of markets, and legal and political systems.

	1	2	3	4	5	6	7	8	9	10
Creditor rights	−0.214 [0.031]**	−0.239 [0.034]**	−0.251 [0.006]***	−0.212 [0.035]**	−0.247 [0.027]**	−0.227 [0.031]**	−0.209 [0.036]**	−0.221 [0.032]**	−0.250 [0.024]**	−0.237 [0.029]**
Depth of credit information	0.623 [0.017]**	0.615 [0.019]**	0.613 [0.016]**	0.601 [0.018]**	0.638 [0.004]***	0.624 [0.013]**	0.598 [0.019]**	0.626 [0.011]**	0.596 [0.026]**	0.611 [0.017]**
Depth of credit information × Creditor rights	0.106	0.109	0.133	0.126	0.162	0.119	0.127	0.114	0.168	0.152
Diversification index	0.235 [0.012]**	0.229 [0.017]**	0.334 [0.019]**	0.332 [0.022]**	0.281 [0.058]*	0.311 [0.042]**	0.313 [0.028]**	0.297 [0.061]*	0.352 [0.012]**	0.337 [0.014]**
Overall capital stringency	0.107 [0.061]*	0.151 [0.036]**	0.120 [0.052]*	0.148 [0.049]**	0.166 [0.048]**	0.121 [0.051]*	0.152 [0.048]**	0.176 [0.047]**	0.192 [0.043]**	0.184 [0.045]**
Financial statement transparency	0.254 [0.041]**	0.226 [0.032]**	0.241 [0.035]**	0.215 [0.052]*	0.311 [0.025]**	0.247 [0.034]**	0.237 [0.037]**	0.264 [0.031]**	0.264 [0.030]**	0.287 [0.029]**
Deposit insurance coverage	−0.124 [0.028]**	−0.197 [0.013]**	−0.159 [0.009]***	−0.157 [0.012]**	−0.145 [0.017]**	−0.157 [0.012]**	−0.158 [0.011]**	−0.149 [0.015]**	−0.155 [0.013]**	−0.153 [0.014]**
Bank size	0.450	0.452	0.437	0.433	0.436	0.426	0.429	0.431	0.433	0.434

Table C1 (continued)

	1	2	3	4	5	6	7	8	9	10
Bank size squared	[0.000]*** –0.015	[0.008]*** –0.017	[0.001]*** –0.016	[0.014]** –0.015	[0.003]*** –0.017	[0.028]** –0.013	[0.024]** –0.014	[0.023]** –0.014	[0.011]** –0.012	[0.008]*** –0.014
Too-big-to-fail	[0.028]** –0.219	[0.023]** –0.268	[0.023]** –0.288	[0.027]** –0.287	[0.022]** –0.274	[0.029]** –0.289	[0.028]** –0.287	[0.027]** –0.281	[0.031]** –0.270	[0.028]** –0.283
HHI	[0.031]** –0.177	[0.034]** –0.124	[0.023]** –0.112	[0.024]** –0.104	[0.031]** –0.126	[0.021]** –0.11	[0.022]** –0.105	[0.027]** –0.124	[0.035]** –0.106	[0.025]** –0.116
Bank growth	[0.103] –0.096	[0.053]* –0.093	[0.063]* –0.071	[0.076]* –0.062	[0.057]* –0.066	[0.071]* –0.061	[0.077]* –0.062	[0.062]* –0.072	[0.073]* –0.063	[0.062]* –0.064
Log GDP per capita	[0.124] 0.194	[0.062]* 0.184	[0.081]* 0.168	[0.163] 0.191	[0.138] 0.258	[0.181] 0.173	[0.176] 0.181	[0.086]* 0.212	[0.188] 0.209	[0.189] 0.201
Log GDP	[0.012]** 0.058	[0.031]** 0.056	[0.011]** 0.096	[0.079]* 0.101	[0.020]** 0.111	[0.102] 0.103	[0.103] 0.100	[0.026]** 0.108	[0.057]* 0.099	[0.042]** 0.108
Inflation	[0.114] –0.081	[0.081]* –0.092	[0.035]** –0.102	[0.049]** –0.109	[0.018]** –0.132	[0.058]* –0.104	[0.034]** –0.105	[0.063]* –0.117	[0.025]** –0.119	[0.027]** –0.116
Creditor rights × Deposit insurance coverage	[0.000]*** –0.036	[0.001]***	[0.000]***	[0.001]***	[0.008]***	[0.000]***	[0.006]***	[0.003]***	[0.000]***	[0.000]***
Min. capital-asset ratio requirement	[0.001]***	0.271								
GDP volatility		[0.042]**	–0.082 [0.000]***							
Control of corruption				0.194 [0.037]**						
Government effectiveness					0.180 [0.188]					
Political stability						0.028 [0.137]				
Quality of regulation							0.129 [0.024]**			
Rule of law								0.079 [0.156]		
Voice and accountability									0.107 [0.259]	
Transparency										0.185 [0.013]**
Constant	1.099 [0.029]**	1.355 [0.032]**	1.694 [0.010]**	1.074 [0.024]**	1.094 [0.037]**	1.253 [0.016]**	1.246 [0.019]**	1.542 [0.030]**	1.763 [0.024]**	1.656 [0.027]**
Observations	2363	2321	2363	2363	2363	2363	2363	2363	2363	2363
Adjusted R ²	0.264	0.260	0.271	0.264	0.263	0.261	0.261	0.262	0.262	0.265
Countries	66	64	66	66	66	66	66	66	66	66

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